

Persuasion and Prejudice: Are South Korean Attitudes Toward Immigration Open to Change?

Steven Denney*

Isa van Dam[†]

Christopher Green[‡]

May 14, 2026

Abstract

How do framing cues shape immigration attitudes in a setting where demographic need is acute but ethnocultural priors remain strong? We examine South Korea, where the government has tied expanded immigration to economic competitiveness and to a deepening fertility crisis. A nationally representative survey experiment (n=1,999) crosses three between-subjects framing arms with a choice-based conjoint on immigrant-admission profiles. The findings separate two levels at which respondents process immigration cues. At the policy level, the pro-immigration growth cue polarizes. It consolidates the status quo on the political center and right, while pushing self-placed progressives sharply toward restriction and more than doubling their share favoring lower immigration. The partisanship cut replicates the pattern only partially. The conservative party consolidates around the status quo, while the progressive party shows the restrictionist backlash without the status-quo consolidation. At the individual level, the headline attribute hierarchies are stable. Origin, religion, language, co-ethnicity, and skill penalties and premiums operate at the magnitudes the cross-national literature documents, independent of the cue. Together, the results identify a separation between cue effects on stated policy preferences, which run through a partisan source-cue channel, and the deeper hierarchies that organize preferences over which immigrants should be admitted, which the cues leave in place.

Keywords: immigration, persuasion, framing, conjoint, status quo activation, South Korea

* Assistant Professor, Leiden University, and corresponding author: s.c.denney@hum.leidenuniv.nl.

[†] Research Assistant, Leiden University

[‡] Assistant Professor, Leiden University

I Introduction

Immigration attitudes in South Korea sit at the intersection of two pressures. The first is the demographic case for liberalization. A total fertility rate of 0.72 in 2023, an aging population, and labor shortages in agriculture, manufacturing, and care work all give policymakers reason to expand channels that the existing visa programs do not cover (Clemens, 2024). The second is a tradition of conditional acceptance and underlying skepticism that has produced a stratified visa regime in which co-ethnic Koreans from advanced economies are privileged over Joseonjok and Goryeoin co-ethnics from China and the former Soviet Union, who are in turn privileged over non-ethnic Korean migrants from middle-income origins (Seol and Skrentny, 2009; Chung, 2020a; Denney and Green, 2021). The country's foreign-born share, even after a decade of growth, sits below five percent of the population.

In democracies, immigration policy reform depends on social support (Levy and Wright, 2020), so understanding whether elite messaging can shift public opinion matters for what reform is politically feasible. Three families of expectations bear on this question. The classical persuasion tradition, organized by the Generalizing Persuasion framework of Druckman (2022) and the competitive-frames program of Chong and Druckman (2007), predicts that strong, broadly applicable frames should shift policy preferences in the direction of the framed argument. A second line of work, anchored by Kustov et al. (2021) and Kustov (2024), finds that immigration preferences are unusually stable and that pro-immigration messaging more reliably raises issue salience than changes direction. A third line, anchored by Arceneaux and Nicholson (2023) on status quo bias and by recent work on the durability of immigration attitudes through the European refugee crisis and the war in Ukraine (Bansak et al., 2023a; Letki et al., 2024), predicts that frames consolidating existing elite consensus activate the status quo position over alternatives.

This study asks whether frames the South Korean government has itself advanced since at least 2018 (In, 2024; Park, 2024) resonate with the public. Specifically, we test whether messages framing immigration as an answer to South Korea's low fertility rate or as essential for maintaining economic growth alter policy preferences and reduce origin- and ethnicity-based penalties at the immigrant-admission stage. These are plausible pro-immigration cues precisely because demographic need is acute and the growth frame has broad elite support. They are also difficult cues because they land in a setting where ethnocultural priors

about belonging remain strong. We therefore ask whether any effects vary with respondents' baseline sociotropic orientation (Hainmueller and Hopkins, 2014; Valentino et al., 2019), and we add subgroup analyses by ideology, partisanship, and sociotropic group because these are the cleavages along which Korean elite messaging would be expected to differentially land.

To address these questions we use a two-part empirical strategy on a nationally representative online panel of 1,999 native-born South Koreans fielded through Qualtrics in January 2024. First, a between-subjects framing experiment evaluates the impact of growth- and fertility-cue messages on stated policy preferences over the size of the foreign-born population. Second, a forced-choice conjoint of immigrant-admission profiles isolates how 11 attributes, including origin, ethnicity, religion, occupation, and Korean language proficiency, structure individual preferences over which immigrants should be admitted. Each respondent evaluates ten paired-profile tasks, with the framing cue reinforced after the fourth task to mitigate fade-out (Bansak et al., 2018).

The findings depart from a uniform "framing fails on immigration" reading in a specific way. Pro-immigration framing in this design operates at the policy level. The cues shift opinions about how much immigration there should be. Preferences over which immigrants to admit remain stable.

First, the growth cue *polarizes* the South Korean public at the policy-preference margin. The modest Maintain lift of 6.6 percentage points in the population conceals opposite-direction effects across self-placed ideology. Centrists and conservatives consolidate Maintain by 11.0 and 11.9 points, while progressives reduce Maintain by 15.7 points and more than double their share selecting *Decrease*, from 9.8% to 22.3%. The Growth $\times$ Centrist interaction on Maintain is +20.6pp ($p=.002$), and the Growth $\times$ Conservative interaction on Maintain is +26.2pp ($p<.001$). The pattern replicates partially on partisanship. PPP identifiers consolidate Maintain by 14 percentage points. DP identifiers (a broader coalition than the Very/Somewhat Progressive ideology bucket) show only the Increase-to-Decrease component of the backlash and hold Maintain flat. We read the cleaner ideology pattern as status quo activation filtered through source politics (Arceneaux and Nicholson, 2023; Nicholson, 2012; Berinsky, 2009). The pattern is inconsistent with the parallel-updating consensus (Coppock, 2023; Wood and Porter, 2019) and with classical competitive-frames persuasion (Druckman, 2022; Chong and Druckman, 2007) as uniform-

population mechanisms. The fertility cue produces no comparable polarization.

Second, looking across the conjoint design, the headline attribute hierarchies are stable. The China origin penalty (about 10 percentage points), the Muslim religion penalty (about 14 percentage points), the Korean-language premium (a 22-point range), the co-ethnic premium, and the skill premium all operate at the magnitudes the prior conjoint literature documents (Bansak et al., 2016; Denney and Green, 2021; Fraser and Cheng, 2024), independent of the cue. An exploratory across-attribute presentation in Section 4.3 identifies one corner where the fertility cue changes estimates. The work-history filter softens. The Agricultural Worker level falls 3.1pp under the fertility cue ($p=.026$). The No-work-history level rises 2.4pp ($p=.035$). The unemployment penalty narrows particularly on ethnic-Korean profiles. The magnitudes are small relative to the deep penalties they sit alongside and do not survive a multiple-comparisons correction across the seventy-plus attribute-level deltas the design produces. We report them as a candidate for further work.

The findings together push debates about persuasion and prejudice in two related directions. The polarization result identifies a boundary condition on the parallel-updating consensus. When pro-immigration framing is part of the elite consensus, source-cue effects can dominate informational ones, producing opposite-direction shifts across the political spectrum. The conjoint-stability result complements this with a clean separation between the policy and the individual levels of the immigration question. Cues shift what respondents say they want as a quantity. They leave candidate sorting largely unchanged. The two findings point to a layered account in which the same cue operates at one level and leaves another stable.

2 South Korean Immigration in Review

South Korea is a textbook case of what migration scholars term a *new immigration country*, a state that transitioned from net emigration to net immigration only in the late twentieth century, lacks the deep policy infrastructure of its Western European counterparts, and confronts the politics of incorporation under conditions of rapid demographic decline (Chung, 2020b; Liu-Farrer, 2020). The category was

originally applied to southern Europe, where Italy, Spain, Portugal, and Greece shifted from sending to receiving countries between the late 1970s and 1990s. The framework now travels well to East Asia, where Japan, South Korea, and Taiwan share late entry into the global migration system, ethnonational ideologies of nationhood, and labor shortages produced by sustained sub-replacement fertility (Chung, 2020b; Maeda and Strausz, 2025). The South Korean case is on the steeper end of the curve. The foreign-born share has climbed from less than half a percent in 1990 to roughly five percent in 2024, with registered foreign residents exceeding two million (Ministry of Justice, 2024), even as total fertility has collapsed to 0.72.

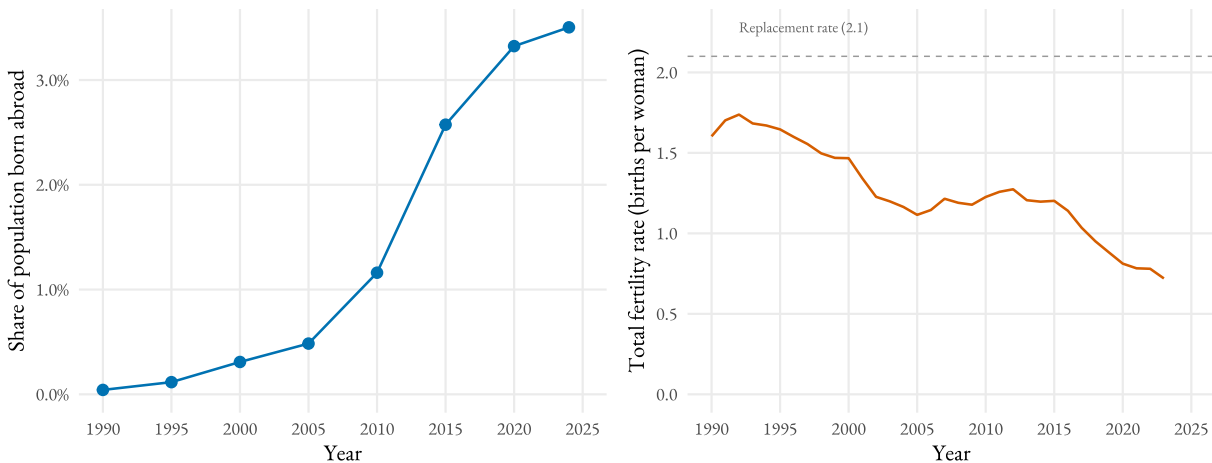


Figure 1: South Korea’s foreign-born share (left) and total fertility rate (right), 1990-2024.

Note: Foreign-born share is the share of the resident population born in another country, from UN DESA’s International Migrant Stock estimates compiled by Our World in Data. Total fertility rate is the average number of births per woman, from the UN World Population Prospects (2024 revision) compiled by Our World in Data. Both series begin in 1990. Migrant-stock data are reported at five-year intervals (1990-2020) plus the latest 2024 estimate. The dashed line marks the population-replacement rate of 2.1 births per woman.

Most foreign residents in South Korea come from neighboring Asian countries, with Chinese nationals, including ethnic Koreans (*Joseonjok*), comprising the largest group at over 900,000. Following China, the most common origins are Vietnam (approximately 270,000), Thailand (202,000), the United States (160,000),¹ Uzbekistan (87,000), and the Philippines (64,000) (Ministry of Justice, 2024).

¹American figures include US military personnel (approximately 28,500).

The legal architecture that channels this growth produces an explicit hierarchy of admission. The Employment Permit System (EPS), introduced in 2004, brings in low-wage workers from fifteen designated Asian countries on temporary, employer-tied contracts, while the Overseas Korean Act (OKA) and the F-4 visa regime offer extensive labor-market and residency rights to ethnic Koreans abroad. The OKA initially excluded co-ethnics from China and the former Soviet Union and prioritized Korean Americans, a structural choice Seol and Skrentny (2009) calls *hierarchical nationhood*, in which bloodline and class are bundled and co-ethnicity alone does not confer equal standing. Two decades on, this hierarchy continues to shape policy and lived experience. Joseonjok and Goryeoin migrants face a narrower set of social and economic opportunities than co-ethnics from advanced economies and consistently report higher rates of perceived discrimination (Song, 2018, 2023; Chung, 2020a). Yu (2023) extends the argument with ethnographic evidence from state-sponsored integration centers, showing that frontline staff sort migrants along an explicit class-and-nationality grid that privileges Western, professional, and ethnically Korean profiles and disadvantages Southeast Asian and Joseonjok workers.

Public-opinion work since 2020 confirms that these institutional hierarchies are mirrored at the individual level. Denney and Green (2021) use a conjoint design to show a stable preference ordering that runs from co-ethnic, high-skill, high-income origins down through Southeast Asian and Chinese profiles. Denney and Green (2024) extend the analysis to integration preferences and find that even when respondents are evaluating co-ethnic migrants, support is conditional on signals of cultural and economic distance from the host society. Gouda and Song (2024) document that economic, cultural, and demographic factors all contribute to attitudes but that the cultural dimension carries disproportionate weight. Choi et al. (2024) adds a useful boundary condition. Respondents in regions experiencing the sharpest population decline are more open to multicultural immigration policy, suggesting that demographic pressure can shift attitudes where it is felt locally rather than as an abstract national figure. Kim and Park (2023) report that Koreans simultaneously acknowledge the economic case for immigration and worry about its social consequences, an ambivalence that helps explain why salient frames seldom shift policy preferences in either direction.

Policy discourse under recent administrations has tracked the same ambivalence. The Moon Jae-in

government (2017-2022) extended visa programs for high-skilled workers and integration support for marriage-migrant families. Yoon Suk-yeol (2022-2025) tied immigration more explicitly to demographic and labor-market necessity but produced limited reform (Jung, 2023). The Lee Jae-myung administration, sworn in in June 2025, has been quiet on the issue. Migrant policy was largely absent from the 2025 campaign (Korea Times, 2025), and Lee's first-year policy agenda has tilted toward restriction. The E-9 quota for low-skilled foreign workers fell from 165,000 in 2024 to 80,000 for 2026, a forty-percent cut justified in domestic-wage-protection terms (Korea Herald, 2025). The silence is itself informative. It indicates that even a progressive government facing the deepest demographic shock in the OECD has not judged the political return on a pro-immigration platform to be positive, which is a useful contextual data point for the framing-experiment results that follow.

2.1 Persuasion, Framing, and Pathways from Cues to Policy Preferences

Whether elite cues can shift public opinion on immigration is among the longest-running questions in the comparative study of attitudes, and the answer has shifted across three generations of research. We adopt the Generalizing Persuasion framework of Druckman (2022) as the organizing scheme. The framework decomposes any persuasion attempt into actors, treatments, outcomes, and settings, and it allows us to be explicit about what kind of persuasion our growth-and-fertility cues attempt, why they differ from the partisan or elite-source cues that dominate the U.S. literature, and which family of empirical predictions we should evaluate. The treatment is a sociotropic policy frame. The receivers are the South Korean general population. The outcome is a stated policy preference over the size of the foreign-born population, and the setting is one in which the government has itself been the loudest sponsor of the frames in question.

The starting point in any modern account of immigration framing is the competitive-frames program of Chong and Druckman (2007) and Chong and Druckman (2013). They show that strong frames typically dominate weak ones, that competition between frames tends to attenuate but not erase persuasion, and that the timing and repetition of counterframes matter for whether an initial frame survives. The critical finding for our purposes is that credibility, applicability, and resonance do the persuasive work in low-partisanship settings. In a context like ours, where the two frames we manipulate are both currently

advanced by the South Korean government and are difficult to map onto a left-right cleavage, the Chong-Druckman model predicts that they should function much like the strong, broadly applicable frames in their experiments.

A first generation of immigration-framing studies took the persuasion premise at face value and asked whether economic, cultural, or humanitarian frames shifted policy preferences. Brader et al. (2008) showed that messages emphasizing immigrants' negative economic and social impact increased restrictionist preferences, especially when paired with cues about Latino origin, with anxiety serving as the mediating mechanism. Bansak et al. (2016), in a fifteen-country conjoint of asylum-seeker profiles, found that respondents reward economic potential, sympathetic biography, and Christian (rather than Muslim) origin. These two studies anchor what we call the *frame-as-preference-shift* tradition. Cues change beliefs about deservingness, threat, or contribution, which then change stated policy preferences. Translated into our setting, the expectation is straightforward. The government's pro-immigration sociotropic frames, which emphasize the contribution of the foreign-born to the labor force and to demographic stability, ought to shift South Koreans toward greater openness on immigration totals.

A second generation of work has cast doubt on this premise. Kustov et al. (2021), using nine panel datasets, document that immigration attitudes are unusually stable over time and survive substantial economic and political shocks, including the 2015 European refugee crisis. Dennison et al. (2023) extend this finding through the COVID-19 period. Policy preferences changed little during the pandemic, while the salience of the issue dropped substantially. Hopkins et al. (2019) and Grigorieff et al. (2020), using corrective-information designs, show that when respondents are told the actual size of the foreign-born population, beliefs update toward accuracy but policy preferences scarcely budge. In Switzerland, Bechtel et al. (2015) find that on the contested 2014 mass-immigration referendum, frames produced no discernible direct effect on the policy positions of voters. The cumulative implication is that, on a salient and contested issue like immigration, most people revise their attention to the question more readily than their preference. In settings where cultural or ethnonational priors are already strong, sociotropic cues may therefore raise the issue without changing the underlying direction of opinion.

Kustov (2024) converts this implication into a positive theoretical claim. He argues that pro-immigration

messaging in particular should be evaluated by whether it raises *issue importance* among existing supporters. In a U.S. survey experiment of $N=3,450$, an information treatment about the national benefits of expanding legal immigration leaves opposition unchanged but increases the priority pro-immigration respondents place on the issue. Jin (2025) clarifies the picture by showing that whether issue salience translates into anti-immigrant attitudes depends on whether elite competition is structured around the issue. In the United Kingdom, where parties are polarized over immigration, increased salience strengthens restrictionist sentiment. In Germany, where elite consensus is more welcoming, salience increases without a corresponding restrictionist turn. South Korea sits closer to the German pole than the British one in the framing offered by sitting governments since at least 2018.

A third literature qualifies both of these accounts by emphasizing status quo bias and the boundaries of elite manipulation. Arceneaux and Nicholson (2023) argue that many political preferences are anchored in the existing policy baseline, making elite cues more likely to activate the status quo than to produce large directional change. The implication is especially relevant for immigration, where the current level of admission is cognitively available and where policy change can be framed as a risk to the existing social order. Coppock (2023) and Wood and Porter (2019) show across hundreds of issues that the predicted "backfire" or "boomerang" response to disliked information is rare, with most respondents updating, however slightly, in the direction the information argues. The early backfire results of Nyhan and Reifler (2010) did not, on replication, generalize. This rules out the strong version of the boomerang interpretation. A pro-immigration cue should not provoke uniform hostile reaction among receivers who already oppose immigration. Helbling et al. (2024) adds a complementary point about conditionality. Respondents balance numbers, selectivity, and rights against each other, and stated preferences shift when the question is framed as a tradeoff against an existing baseline. Why this conditionality should be especially constrained in the South Korean case, where sociotropic reasoning passes through an unusually cohesive set of ethnocultural priors, is developed in Appendix E.

Three expectations follow.

Expectation 1 *Druckman-style persuasion. Growth and fertility cues will shift South Koreans in the liberalizing direction, increasing the share who select Increase and decreasing the share who select Decrease.*

Expectation 2 *Kustov-style salience.* Cues will leave the directional outcome flat while raising the importance respondents place on immigration policy. Under this expectation, a null on the policy-preference DV is informative because the expected change is attention, not direction.

Expectation 3 *Status quo activation.* Cues will consolidate support for maintaining current immigration levels, especially among respondents whose priors are compatible with the executive's sociotropic argument. Under this expectation, pro-immigration messaging activates the existing policy baseline and limits generalized liberalization.

These expectations make distinct predictions on the directional outcome and are jointly testable at the population level. The Korean case is well suited to a direct test of them because the growth and fertility frames we evaluate are actual frames the executive has used in published policy documents and in major-newspaper interviews, including the 2024 Korea Chamber of Commerce and Industry exchange that explicitly tied falling fertility to a need to lower "immigration barriers" (In, 2024; Park, 2024). As we show below, the pattern we estimate is closest to status quo activation but is politically asymmetric. The executive-branded growth message consolidates Maintain among groups whose priors are broadly compatible with the cue and is rejected by groups that read it as out-party-aligned, even when its content matches their priors. We develop that source-cue mechanism (Nicholson, 2012; Berinsky, 2009; Lenz, 2012) in Section 4 and the conclusion.

Both expectations address the same dependent variable, how much immigration the public says it wants. The conjoint half of the design tests a complementary question, whether the same cues alter who the public is willing to admit. Because pro-immigration framing might in principle operate at the policy aggregate, at one or more specific attributes (skill, origin, language, work history), at none of these, or differently at each, the conjoint analysis we report in Section 4.3 is exploratory. We present marginal means and Control-referenced deltas across the six substantively-interesting attributes in the design and identify which attributes, if any, the cues shift. This is the approach the conjoint methodology literature recommends when the location of cue effects is not specified in advance (Hainmueller et al., 2014; Leeper et al., 2020), and the across-attribute presentation makes any selective reporting visible by construction.

3 Data and Methodology

This study uses a two-part empirical design to examine whether persuasion cues centered on demographic decline and economic growth shift South Korean public support for more immigration and reduce origin- and ethnicity-based penalties at the immigrant-admission stage. First, a between-subjects framing experiment tests whether growth- and fertility-cue messages alter immigration policy preferences. Second, a forced-choice conjoint analysis evaluates how the cues shape preferences over hypothetical immigrant applicants on attributes of origin, ethnicity, occupation, religion, language, and prior history of integration. Both components draw on a nationally representative online panel of 1,999 native-born South Koreans fielded through Qualtrics in January 2024. Quotas were set on age, sex, education, and region following the most recent Statistics Korea population parameters. Post-stratification weights were computed by raking on the same dimensions. Appendix A in the Supplementary Information document provides a full sample summary.

Participants were randomly assigned to one of three introductory conditions, illustrated as cards in Figure 2.

Control	Economic Growth Cue	Birthrate Cue
Next, you will be asked your opinion about immigration. You will rate the immigrant profile as if you were an immigration officer.	The government believes that the only way to maintain economic competitiveness and a high standard of living is to accept more immigrants, regardless of background.	South Korea has one of the lowest fertility rates among advanced industrial countries. The government believes the only way to solve the economic and social problems caused by the low birth rate is to accept more immigrants, regardless of background.

Figure 2: Persuasion-cue intervention in the three between-subjects framing arms.

Note: Verbatim English translations of the Korean prompts. Respondents were assigned to one of the three arms with equal probability at survey entry, with realized per-arm sizes 664 / 637 / 698 in the analytic sample (Control / Growth / Birthrate). The cue is reinforced after the fourth conjoint task to mitigate fade-out.

After exposure to one of the three messages, respondents were asked whether South Korea's immigration levels should be *increased*, *decreased*, or *remain the same*. This single-choice question is the framing-experiment dependent variable. We note that the growth and fertility messages both close with the phrase "regardless of background" (*baebyeong-e gwangyeopsi*). The phrase is intentionally vague and is anti-discrimination in direction. It pushes respondents toward equal treatment across origins. We return to this design property in the discussion of the conjoint results, because it makes the framing manipulation conservative against the origin-discrimination patterns we report.

Arm assignment was concealed by design. The Qualtrics random-integer flow element runs at the start of the survey, and the resulting value is never displayed to the respondent. Respondents are blinded to their condition, with each respondent seeing only the cue text relevant to their arm. The analysts are not blinded, because arm labels appear in the post-fielding data extract. All reported analyses are intention-to-treat. Every randomized respondent is analyzed in their assigned arm regardless of attention-check performance, response time, or any other post-treatment behavior. Attrition by arm is structurally zero

in this single-session online design conditional on survey completion. The five respondents dropped between the 2,004 raw finished responses and the 1,999 analytic sample are missing on at least one raking dimension (four on age, one on region), distributed across arms as Control = 2, Growth = 3, Birthrate = 0, and uncorrelated with the framing manipulation, the conjoint task, or the attention check. Appendix A reports per-arm Ns at every flow stage and a balance table across all pre-treatment covariates.

To examine the determinants of admission preferences we used a forced-choice conjoint analysis, a method now standard in immigration-attitude research (Hainmueller and Hopkins, 2015; Denney and Green, 2021; Bansak et al., 2016; Fraser and Cheng, 2024). Respondents were asked to imagine themselves as an immigration officer reviewing the profiles of two prospective immigrants and to select the one they believed should be admitted. They evaluated ten paired-profile tasks, choosing between the two profiles in each task. Each profile is randomized along eleven attributes: country of origin, ethnicity, occupation, Korean language proficiency, religion, sex, marital status, age, years in Korea since entry, employment history, and Korea visit history. (A twelfth attribute, social experience in Korea, was randomized in a half-sample subset of tasks. We use it for sensitivity analysis only.) Table 1 shows the attributes and levels.

Attribute	Levels	Rationale
Country of origin*	United States (ref.) China Indonesia Brazil	Spans cultural, economic, and political distance from the South Korean baseline.
Ethnicity*	Ethnic Korean (ref.) Non-ethnic Korean	Tests the co-ethnic preference and the Overseas Korean Act lineage.
Occupation	Agricultural Worker (ref.) Unemployed Office Worker Computer Programmer	Captures the skill premium and the asymmetry between high-status and stigmatized occupations.
Korean language	Fluent (ref.) Conversational Beginner No Korean	A core integration signal in South Korean public discourse.
Religion	Non-religious (ref.) Christian Buddhist Muslim	Captures religious distance from the plurality non-affiliated baseline. The Muslim level tests anti-Muslim bias documented elsewhere.
Years in Korea	2 years (ref.) 7 years 12 years	Tests whether longer residence attenuates origin and ethnicity penalties at the admission stage.
Marital status	Single (ref.) Married Married with children	Captures family-stability signals relevant to integration.
Profile age	27 (ref.) 36 47 58	Tests age preferences in admission.
Employment history	No prior work (ref.) Worked in home country Worked in Korea	Distinguishes labor-market track records, including in-Korea experience.
Korea visit history	Visited once (ref.) Short-term work visa Previous visa overstay	Captures prior contact with the country, including potentially compromising contact.
Sex	Woman (ref.) Man	Tests gender bias in admission preferences.

Table 1: Conjoint attributes and levels.

Note: Reference levels are marked “(ref.)” Attributes marked with an asterisk (*) are primary quantities of interest. Each respondent evaluated ten paired-profile tasks. The persuasion cue was reinforced after the fourth task.

The principal quantities of interest are origin and ethnicity, together with their interaction with skill (occupation). Country-of-origin levels span cultural, economic, and political distance. The United States is a high-status, high-income baseline. China is the largest actual immigrant source and politically distant. Indonesia is a Muslim-majority Southeast Asian middle-income source. Brazil is a Latin American middle-income comparator. The persuasion cue is reinforced after the fourth profile to mitigate fade-out of the framing manipulation.

We report two complementary quantities, average marginal component effects (AMCEs) and marginal means (MMs). The AMCE estimates the change in a profile’s selection probability when a single attribute changes from a baseline level to a target level, averaging over the joint distribution of the remaining attributes (Hainmueller et al., 2014). The MM reports the average selection probability for a given level. Recent methodological work shows that conditional AMCEs, on which the framing-by-attribute analysis relies, are sensitive to the baseline category and therefore poorly suited for substantive comparisons across subgroups (Leeper et al., 2020; Bansak et al., 2023b). Marginal means are baseline-independent and have a direct probabilistic interpretation. We use AMCEs to characterize main effects relative to a baseline (the United States for origin, Ethnic Korean for ethnicity) and report MMs in parallel for substantive interpretation, particularly for treatment-stratified comparisons. Standard errors are clustered by respondent, and survey weights are applied throughout (Esarey and Menger, 2019).

Appendix D reports the full power and minimum-detectable-effect (MDE) specifications for each estimand class.

4 Findings

Section 4 reports an exploratory pattern in policy preferences that emerged from the across-attribute reanalysis of the rebuilt design. Section 4.2 reports the pooled conjoint preferences, which replicate the cross-national attribute hierarchies the literature documents and serve as the confirmatory descriptive baseline. Section 4.3 returns to a cue-by-attribute analysis across the design.

The findings depart from a uniform "framing fails on immigration" reading in a specific way. The cues

shift opinions about immigration policy. Preferences over which immigrants would be admitted remain stable. On policy preference, the growth cue *polarizes* along ideological lines, consolidating Maintain in the center and on the right while pushing progressives toward Decrease. In the conjoint, origin, ethnicity, religion, language, and skill penalties are all stable across cues at roughly the magnitudes documented in cross-national work. The framing manipulation does not displace the attribute hierarchy on which actual admissions decisions in this design are made.

4.1 The Growth Cue Polarizes Policy Preferences

Figure 3 presents predicted probabilities of support for each immigration-policy option under each framing arm, pooled across ideology. In the control group, 45.4% of respondents favor maintaining current immigration levels, while 38.5% support increases and 16.1% prefer reductions. The growth cue significantly increases support for the status quo by 6.6 percentage points (to 52.0%), drawing primarily from those otherwise inclined toward liberalization, with Increase falling 5.7 points. The fertility cue produces no statistically distinguishable shifts, with support for the status quo rising 1.7 points and Increase falling 2.9 points.

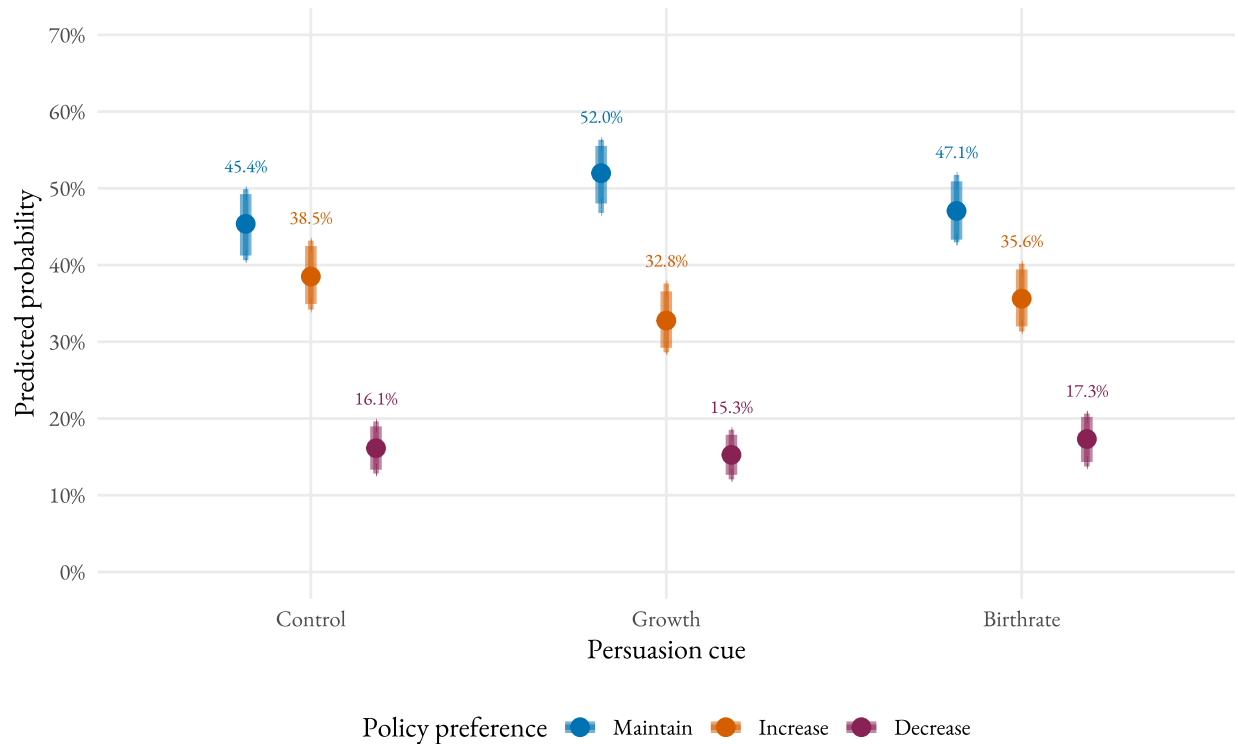


Figure 3: Persuasion cues and immigration-policy preferences (pooled).

Note: Predicted probabilities estimated using multinomial logistic regression with controls for sex, age group, education, region, ideology, and partisanship, and post-stratification survey weights. Error bars show 90% (thick) and 95% (thin) bootstrap confidence intervals. $n = 1,996$ (model-eligible respondents from the raked sample of 1,999).

The pooled Maintain lift bears directly on the status quo expectation. The subgroup polarization is exploratory, because it emerged from the rebuilt analysis rather than from a primary stated test.

The pooled estimate, however, conceals a pattern across political subgroups. Figure 4 disaggregates two ways. Panel A shows the cut by self-placed ideology (1-10 scale, collapsed to Progressive, Centrist, and Conservative). Panel B shows the cut by reported party identification (Democratic Party, Independent, People Power Party). The two cuts show related patterns. On the ideology axis, the growth cue produces opposite-direction shifts across the spectrum. Centrists consolidate Maintain by 11.0 points, from 46.0% in the control arm to 57.0% under the growth cue, with Increase falling 8.4 points. Conservatives consolidate Maintain by 11.9 points (40.2% to 52.1%), with Decrease falling 6.4 points. Progressives shift in the opposite direction. Their Maintain share falls 15.7 points (50.6% to 34.9%), and the share selecting Decrease more

than doubles, from 9.8% to 22.3%, an absolute shift of 12.5 percentage points among the group that should have been the easiest target for pro-immigration persuasion.

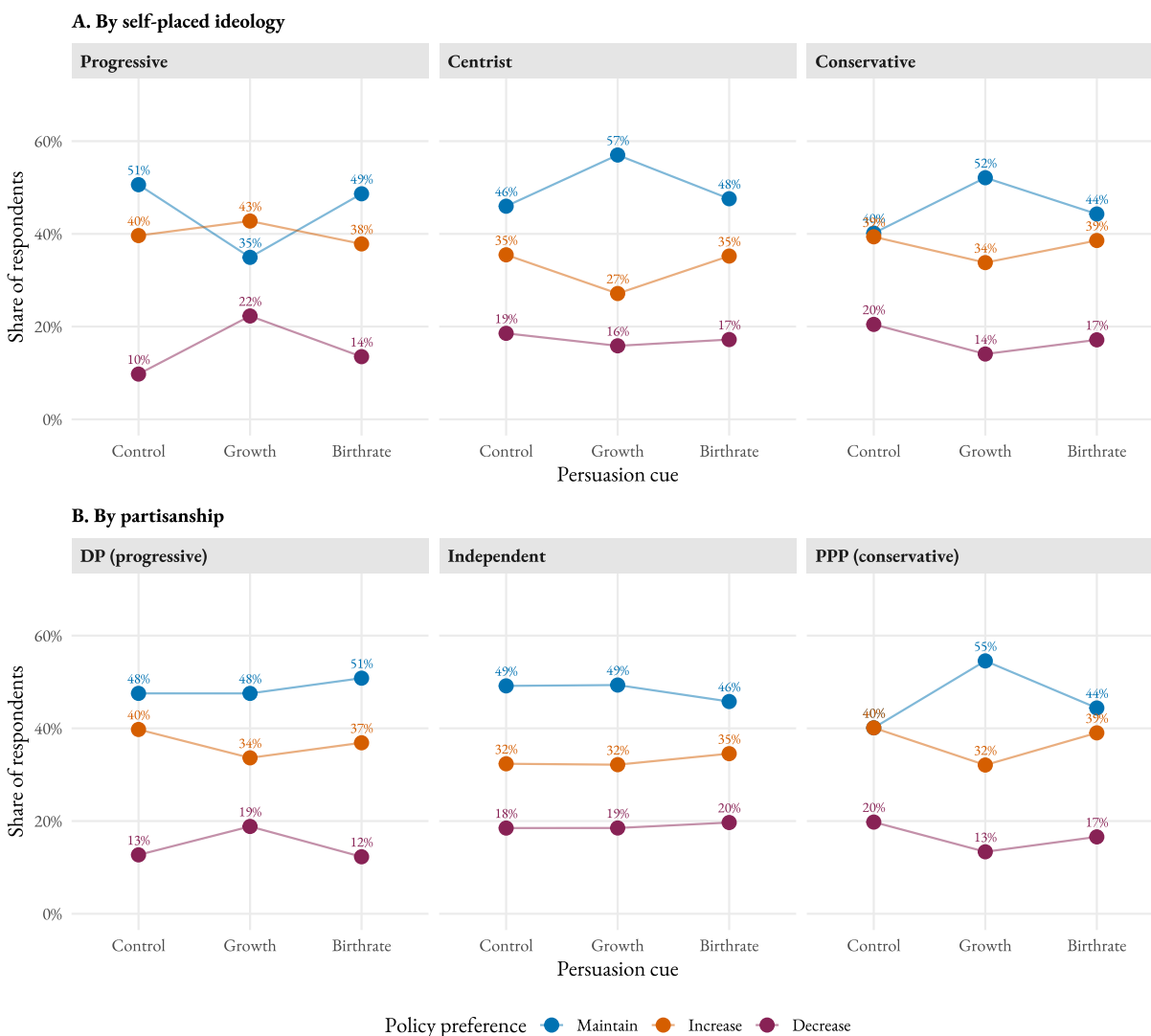


Figure 4: Predicted policy preferences by persuasion cue, disaggregated by political subgroup.

Note: Cell shares within subgroup $\times$ frame. Panel A shows ideology measured pre-treatment on a 1-10 self-placement and collapsed to Progressive (1-4), Centrist (5-6), Conservative (7-10). Within-cell n ranges from 127 to 372. Panel B shows partisanship from a pre-treatment item, collapsed to Democratic Party, Independent (DK/Other/Justice-Green), and People Power Party. Within-cell n ranges from 174 to 249. Error bars are 90% (thick) and 95% (thin) confidence intervals with respondent-clustered standard errors and survey weights.

The growth $\times$ ideology interactions are statistically detectable. In a linear-probability model with respondent-clustered standard errors and demographic controls, Growth $\times$ Centrist on Maintain is +20.6

percentage points ($p=.002$), and Growth $\times$ Conservative on Maintain is +26.2 percentage points ($p<.001$). Decrease shows the mirror pattern. Growth $\times$ Centrist on Decrease is -13.0 pp ($p=.010$). Growth $\times$ Conservative is -15.2 pp ($p=.011$).

The partisanship cut (Panel B) tracks the ideology pattern only partially. The conservative pole replicates cleanly. PPP identifiers consolidate Maintain by 14 percentage points (40.1% to 54.5%) and reduce Decrease by 6 points, with Growth $\times$ PPP on Maintain at +14.0pp ($p=.041$) and Growth $\times$ PPP on Decrease at -11.0 pp ($p=.036$). The progressive pole does not. DP identifiers under the growth cue hold Maintain essentially flat (47.5% to 47.5%) while shifting 6 points from Increase to Decrease. The Maintain decline observed among self-placed progressives is therefore not present among the broader DP coalition. Only the Increase-to-Decrease component of the backlash carries over. We interpret the asymmetry as straightforward measurement. Self-placed Progressives on a 1-10 scale ($n = 515$) are a smaller, ideologically more activated subset than the DP party-ID coalition ($n = 711$). The source-cue rejection operates most clearly on respondents whose stated ideology is far enough left to read the executive cue as out-party-aligned. Among more centrist DP identifiers, the cue is closer to neutral and the Maintain share does not collapse. The fertility cue produces no comparable polarization on either cut.

This pattern fits the status quo expectation in one part of the electorate and departs from it in another. Druckman-style persuasion (Expectation 1) predicts liberalization in the cue's argued direction. We instead observe restrictionism among the ideologically nearest group. Kustov-style salience (Expectation 2) predicts a flat directional outcome and is contradicted by the magnitude of the centrist and conservative shifts on Maintain. Status quo activation (Expectation 3) captures the consolidation among centrists and conservatives. The progressive backlash adds a *source-cue* or *elite-consensus rejection* component of the kind documented by Nicholson (2012) and Berinsky (2009), in which an executive-branded message is rejected by partisans of the out-party even when its content matches their priors. The cue's pro-immigration content is consistent with progressive priors, but the cue is also unmistakably the South Korean executive's framing, which Yoon (2022-2025) and Lee (2025-) have both invoked. Progressives reading the cue as a Yoon/Lee-aligned message reject the policy posture over the content.

The polarization pattern is larger than a noise artifact. The growth cue's 20.6pp Centrist-vs-Progressive

Maintain interaction and the 12.5pp progressive Decrease shift both sit well above the design’s minimum detectable interaction effect (see Appendix D). The pooled fertility-cue effects on Increase, by contrast, are underpowered. The data are consistent with anything from a small liberalizing effect to a small consolidation effect but rule out a large effect in either direction.

For completeness, we also report results by sociotropic baseline orientation (Appendix C, Figure C.1). Among optimists (top quartile of the sociotropic composite, $n = 370$), the control-arm distribution is 31% Maintain, 63% Increase, 5% Decrease. The growth cue lifts Maintain 13 points and reduces Increase 16 points. Among skeptics ($n = 1,006$), the cue lifts Maintain 11 points. The pattern qualitatively matches the ideology pattern. The cue consolidates Maintain among groups whose priors are not far from the cue’s content. Subgroup interactions with treatment in the multinomial logit are not statistically distinguishable from zero at conventional levels, likely reflecting limited statistical power at the within-arm subgroup sample sizes (optimists at roughly 123 per arm).

4.2 Pooled Conjoint Admission Preferences

Before stratifying by framing arm, we summarize the average admission-preference structure that the conjoint produces across all respondents and all framing arms. The full attribute hierarchy of marginal means is reported in the Supplementary Information (Appendix C, Figure C.2). Here we report the largest patterns and use them as the baseline against which the framing-stratified analysis in Section 4.3 is read.

Three attributes carry the largest variation in selection probability. Korean-language fluency spans a 21.7-percentage-point range, from *No Korean* at 36.8% to *Fluent* at 58.5%. Religion spans 14.8 points, with the *Muslim* level at 39.1% well below the *Non-religious* baseline at 53.9%. Country of origin spans 11.8 points, with China at 42.9% and the United States at 54.7%. The next four attributes carry smaller but non-trivial ranges. Occupation spans 10.5pp (Computer Programmer at 55.1% above Unemployed at 44.6%), employment history 7.9pp (prior work in Korea above no work history), ethnicity 7.5pp (Korean vs. non-Korean), and profile age 6.9pp (younger profiles preferred). Marital status, Korea visit history, years in Korea, and profile sex carry the smallest ranges (under 6pp each). Korean fluency on its own is large enough to override origin. A Fluent Chinese profile is selected at 50.8%, beating a No-Korean

American profile at 41.3%.

Two substantive points follow before we stratify by frame. The dominant attribute pattern is integrationist rather than ascriptive. Acquired civic markers (language, work history, time in Korea) carry as much or more weight than the unalterable ones (origin, ethnicity). The largest individual penalties in the design are religion-based (Muslim) and origin-based (China), although the prior literature on hierarchical nationhood would predict the ethnic-Korean / non-ethnic-Korean penalty to dominate at the public-attitude level (Seol and Skrentny, 2009; Denney and Green, 2021). Both patterns matter for interpreting the framing manipulation. Origin and religion penalties of 10 to 14 percentage points are an order of magnitude larger than the framing-experiment shifts on policy preference, which means a few-point framing effect on a non-trivial subset of cells is substantively interpretable but does not displace the underlying hierarchy.

4.3 Conjoint Admission Preferences Across Attributes

We look across the conjoint design to see where, if anywhere, the cues produce detectable changes. Figure 5 presents the result in two columns. Panel A reports marginal means by framing arm for the six substantively-interesting attributes (origin, ethnicity, religion, Korean language, occupation, employment history). The three points per row are Control, Growth, and Birthrate, with 90 and 95 percent confidence intervals. Panel B reports the same data differenced against Control, so that each row's Growth—Control and Birthrate—Control deltas appear next to the level it pertains to. The remaining five attributes (profile sex, marital status, profile age, years in Korea, Korea visit history) are reported in the Supplementary Information.

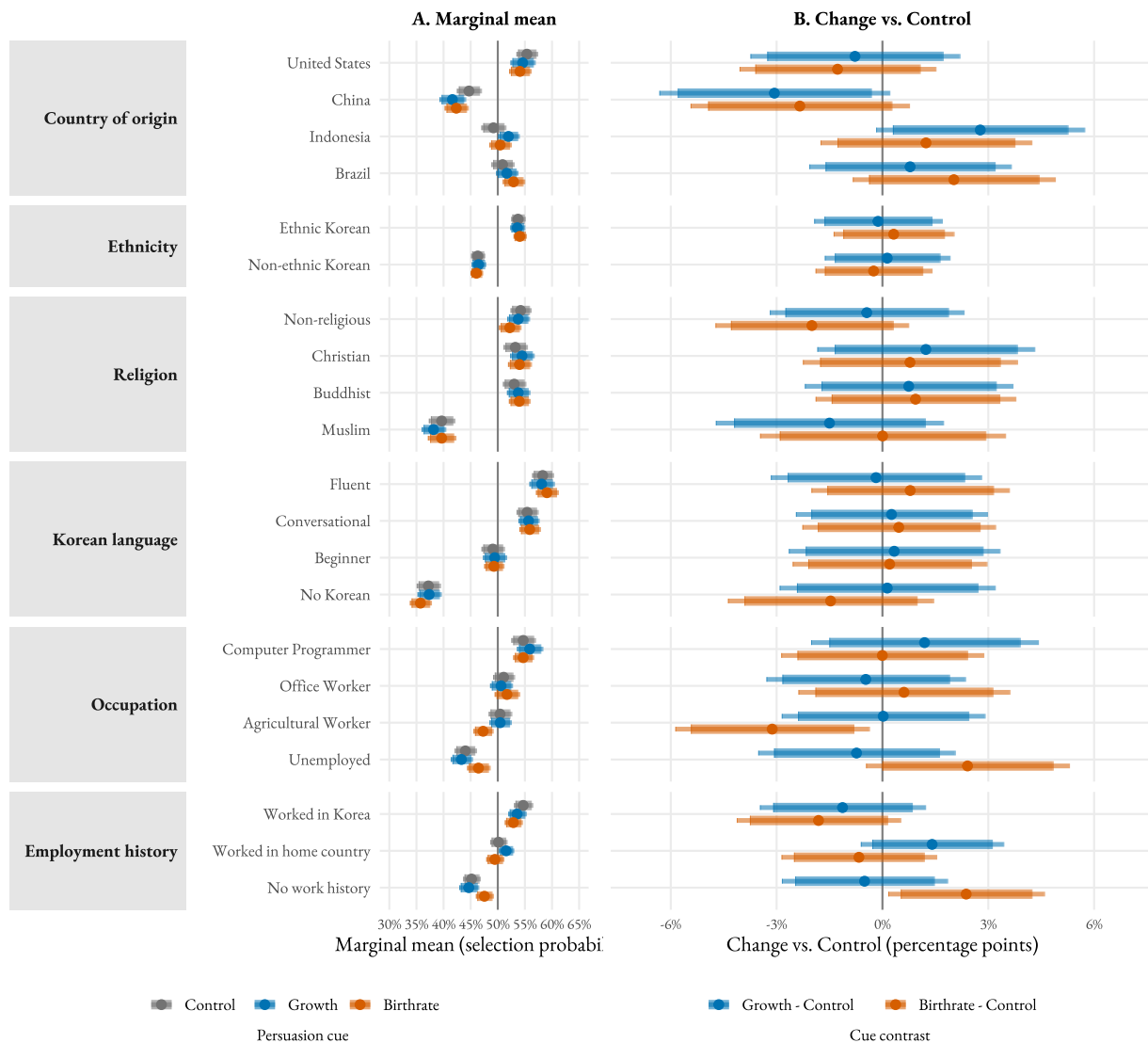


Figure 5: Conjoint admission preferences across framing arms. Marginal means and cue deltas.

Note: Panel A reports marginal means by persuasion cue for each level of the six substantively-interesting attributes. Panel B reports the change in marginal mean relative to Control under each cue, on the same rows. Three-point dodge in Panel A shows Control (gray), Growth (blue), Birthrate (orange). Two-point dodge in Panel B shows Growth—Control (blue), Birthrate—Control (orange). Error bars are 90% (thick) and 95% (thin) confidence intervals with respondent-clustered standard errors and survey weights. Sample $n = 1,999$. Reference line in Panel A is at chance selection (0.5). Reference line in Panel B is at zero.

The headline attribute hierarchies are stable. The largest penalties in the design operate at the same magnitudes under every framing arm. Chinese profiles are selected at about 43% against 55% for U.S. profiles, an 11.8pp range. Muslim profiles sit at about 39% against 54% for non-religious, a 14.8pp range.

No Korean sits at 37% against Fluent at 59%, a 21.7pp range. The Control, Growth, and Birthrate point estimates overlap across the confidence intervals in Panel A for all three. The ethnicity penalty (a 7.5pp Korean vs. non-Korean gap) shows the same property. These are the patterns the prior cross-national conjoint literature documents (Bansak et al., 2016; Helbling and Traunmüller, 2020; Fraser and Cheng, 2024; Denney and Green, 2021). The framing manipulation does not displace them. Cue effects on origin AMCEs change by no more than 2.5 percentage points across the three arms, below the design's MDE for treatment-by-origin interactions. Appendix D reports the MDE at about 4.9pp.

Two attributes change under the fertility cue. Panel B makes the location of the changes legible. On occupation and employment history, the Birthrate—Control deltas (orange) sit visibly off zero on three rows, all in directions consistent with a relaxation of the work-history filter. The Agricultural Worker cell falls 3.1 percentage points ($p=.026$). The No-work-history cell rises 2.4 points ($p=.035$). The Unemployed cell rises 2.4 points ($p=.10$). The Worked-in-Korea cell falls 1.8 points ($p=.13$). The growth cue produces no detectable change on any attribute. Two qualifications matter before interpretation. First, these are the only attributes in the design with pooled cue deltas reaching the conventional significance threshold, and the pattern does not survive a standard correction for the more than seventy attribute-level comparisons in Figure 5. Second, the magnitudes (2 to 3 percentage points pooled) are an order of magnitude below the deep penalties on origin and religion that they sit alongside. We treat the pattern as a small but coherent exploratory finding.

Our reading of this pattern is conservative. Under the fertility cue, respondents appear to soften the work-history filter on profiles whose other attributes are not otherwise disqualifying. A profile coded as Unemployed or with no prior work history becomes slightly more selectable under a frame that emphasizes the need for more residents in a country with collapsing fertility. A profile already deeply embedded in the Korean labor market (Worked in Korea) becomes slightly less so. The pattern is sharpest on a three-way interaction discussed in the SI. Under the fertility cue, the unemployment penalty for ethnic-Korean profiles (the gap between Computer Programmer and Unemployed) narrows from 12.2 to 4.3 percentage points, a 5.6pp lift on the Unemployed Ethnic-Korean cell that is statistically distinguishable from zero in a respondent-clustered linear-probability model with all other attributes controlled ($p<.001$). We do not

theorize this further. It is the kind of cue-by-attribute pattern that warrants follow-up in a design built explicitly to test it.

Taken with the polarization result on policy preference (Section 4, subsection 4.1), the conjoint result identifies a separation between two levels at which the cues operate. Pro-immigration framing in this design shifts opinions about how much immigration there should be, strongly and in opposite directions across the ideological spectrum, while leaving the headline attribute hierarchies that organize *which* immigrants respondents would admit in place. The work-history filter under the fertility cue points to a specific candidate for further work rather than a substantive answer in this study.

5 Conclusion

This study examined whether pro-immigration messages framed around economic growth and demographic decline could shift South Korean attitudes toward immigration policy and toward immigrant-admission criteria. The two questions sit at different levels, and the rebuilt analysis returns different answers at each. At the policy level, the framing manipulation produces large, partisan-coded shifts. At the individual level, headline admission preferences remain stable.

The growth cue produces a 6.6-point lift in support for maintaining current immigration levels at the population average, but the average conceals opposite-direction subgroup effects. Centrists and conservatives consolidate Maintain by 11 to 12 percentage points. Growth $\times$ Centrist on Maintain is +20.6pp ($p=.002$), and Growth $\times$ Conservative on Maintain is +26.2pp ($p<.001$). Self-placed progressives, by contrast, more than double their share selecting *Decrease* (9.8% to 22.3%) and reduce their Maintain share by 16 points. On the partisanship axis, PPP identifiers consolidate Maintain by 14 points. DP identifiers, a broader coalition than the Very/Somewhat Progressive ideology bucket, show only the Increase-to-Decrease component of the backlash and hold Maintain flat. We read the cleaner ideology pattern as status quo activation filtered through source politics. The cue consolidates respondents whose priors are compatible with the executive's sociotropic argument and triggers rejection among respondents whose stated ideology is far enough left to read the cue as out-party-aligned. The partial partisanship

replication suggests that the same mechanism dilutes across the broader party coalition.

The same cues leave the headline preferences over individual immigrants stable. Looking across all six substantively-interesting attributes the conjoint manipulates, the origin penalty for Chinese profiles (about 10 percentage points relative to U.S. profiles), the Muslim religion penalty (about 14 percentage points), the Korean-language premium (a 22-point range), the co-ethnicity premium, and the skill premium operate at the magnitudes the prior cross-national conjoint literature documents (Bansak et al., 2016; Denney and Green, 2021; Fraser and Cheng, 2024), independent of the cue. Two attributes do show cue-linked changes under the fertility frame. The Agricultural Worker level falls 3.1 percentage points and the No-work-history level rises 2.4 points, with a parallel narrowing of the unemployment penalty on ethnic-Korean profiles in particular. The magnitudes are small relative to the 10-14pp origin and religion penalties, and the pattern does not survive a multiple-comparisons correction across the seventy-plus attribute-level deltas the design produces. We report it as an exploratory observation. The work-history filter is the location at which a future, more focused experiment could test the framing-effects question we cannot answer here.

Two implications for policy and political communication follow. In elite-consensus contexts where pro-immigration framing is already the official line, generic sociotropic messaging can consolidate the status quo among some groups and polarize at the policy-attitude margin among others. The growth cue's progressive backlash is a cautionary case for governments considering a national pro-immigration campaign. Specific arguments tied to identifiable beneficiaries and source-diverse delivery appear in the comparative literature to be more effective (Kustov, 2024; Hopkins et al., 2026; Nicholson, 2012). The conjoint-stability result suggests that the cue works at the level of partisan signaling about the policy posture, with little change in who respondents would admit. Even where the cue shifts stated policy preferences by double-digit margins, it leaves the underlying penalties on China, on Muslim profiles, and on low-skill candidates in place.

Theoretically, the findings push debates about persuasion and prejudice in two related directions. They identify a boundary condition on the parallel-updating consensus. In a setting where ethnocultural priors are strong and demographic need is severe, pro-immigration framing activates the status quo among respondents for whom the executive cue is compatible with prior commitments and produces source-cue

rejection among those who read the message through out-party politics. The findings also specify a separation between the policy and individual levels at which respondents process immigration cues. The framing manipulation shifts what respondents say they want as a quantity. It leaves candidate sorting largely unchanged. This complicates both the Kustov salience-not-direction account (Kustov, 2024) and the standard "no effect" reading of immigration framing nulls, and it suggests that future work should specify which level a cue is expected to operate on before testing it.

Our framing manipulation is a single-paragraph cue rather than a full media-environment manipulation, so effect sizes are bounded above by the strength of the cue. The polarization findings emerge inside a single-paragraph cue, and the comparative literature suggests real-world media-environment manipulation should amplify them. The work-history finding identified in Section 4.3 is a natural next target. A design built specifically to test whether a fertility frame relaxes admission preferences over candidates with weaker labor-market histories, with a larger sample on the cells in question and an a priori restriction to that attribute family, would establish whether the small pooled deltas we observe are a substantive cue effect or noise. An additional avenue for future research concerns the intersection of demographic decline, national security, and immigration. In South Korea, population aging and low fertility are also security concerns in a state that maintains compulsory military service under persistent threat from North Korea. Whether security-oriented cues, linking immigration to sustaining national defense capacity, affect public opinion in similarly polarizing ways is an important extension our design does not address (Bigo, 2002).

Several further limitations frame the inferences we draw. The study is a single country, single fielding (January 2024), and a panel-based online sample fielded through Qualtrics. Generalization across countries with different ethnonational baselines, across panel and probability samples, and across time, including post-2024 changes in partisan composition, requires replication. The design also carries no manipulation check, so we cannot verify that respondents in the Growth and Birthrate arms retained the cue at the moment of conjoint choice. Reinforcing the cue after the fourth task mitigates fade-out but is not itself validated. The two cues we test are the ones the South Korean executive has actually advanced in published policy documents, a strength for external validity and a constraint on causal generality. Other plausible cues, such as a security frame, a humanitarian frame, or an economic-redistribution frame, are absent

from the design and may produce different patterns. The polarization pattern we report in Section 4 emerged from the across-attribute analysis of the rebuilt design, and we offer it as a candidate finding worth replication in a future fielding. The work-history finding in Section 4.3 is exploratory at a magnitude (2 to 3 percentage points pooled) small enough that confirmation in a future fielding is warranted.

6 Funding

This work was supported by the Academy of Korean Studies under Grant AKS-2023-R-018.

7 References

- Arceneaux, K. and Nicholson, S. P. (2023). Anchoring political preferences: The psychological foundations of status quo bias and the boundaries of elite manipulation. *Political Behavior*, 45(4):1737–1758.
- Bansak, K., Hainmueller, J., and Hangartner, D. (2016). How economic, humanitarian, and religious concerns shape European attitudes toward asylum seekers. *Science*, 354(6309):217–222.
- Bansak, K., Hainmueller, J., and Hangartner, D. (2023a). Europeans’ support for refugees of varying background is stable over time. *Nature*, 620:849–854.
- Bansak, K., Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2018). The number of choice tasks and survey satisficing in conjoint experiments. *Political Analysis*, 26(1):112–119.
- Bansak, K., Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2023b). Using conjoint experiments to analyze election outcomes: The essential role of the average marginal component effect. *Political Analysis*, 31(4):500–518.
- Bechtel, M. M., Hainmueller, J., Hangartner, D., and Helbling, M. (2015). Reality bites: The limits of framing effects for salient and contested policy issues. *Political Science Research and Methods*, 3(3):683–695.

- Berinsky, A. J. (2009). *In Time of War: Understanding American Public Opinion from World War II to Iraq*. University of Chicago Press, Chicago, IL.
- Bigo, D. (2002). Security and immigration: Toward a critique of the governmentality of unease. *Alternatives: Global, Local, Political*, 27(1_suppl):63–92.
- Brader, T., Valentino, N. A., and Suhay, E. (2008). What triggers public opposition to immigration? anxiety, group cues, and immigration threat. *American Journal of Political Science*, 52(4):959–978.
- Choi, S., Song, J., Kwon, D., and Kim, B. H. S. (2024). Population decline and public attitudes toward multicultural immigration policies in South Korea. *Population, Space and Place*, 30(6):e2788.
- Chong, D. and Druckman, J. N. (2007). Framing public opinion in competitive democracies. *American Political Science Review*, 101(4):637–655.
- Chong, D. and Druckman, J. N. (2013). Counterframing effects. *Journal of Politics*, 75(1):1–16.
- Chung, E. A. (2020a). Creating hierarchies of noncitizens: Race, gender, and visa categories in south korea. *Journal of Ethnic and Migration Studies*, 46(12):2497–2514.
- Chung, E. A. (2020b). *Immigrant Incorporation in East Asian Democracies*. Cambridge University Press, Cambridge.
- Clemens, M. A. (2024). Migration or stagnation: Aging and economic growth in korea today, the world tomorrow. *SSRN Electronic Journal*.
- Coppock, A. (2023). *Persuasion in Parallel: How Information Changes Minds about Politics*. Chicago Studies in American Politics. University of Chicago Press, Chicago, IL.
- Denney, S. and Green, C. (2021). Who should be admitted? conjoint analysis of south korean attitudes toward immigrants. *Ethnicities*, 21(1):120–145.
- Denney, S. and Green, C. (2024). Public attitudes towards co-ethnic migrant integration: Evidence from South Korea. *Journal of Ethnic and Migration Studies*, 50(8):1998–2022.

- Dennison, J., Kustov, A., and Geddes, A. (2023). Public attitudes to immigration in the aftermath of COVID-19: Little change in policy preferences, big drops in issue salience. *International Migration Review*, 57(2):557–577.
- Druckman, J. N. (2022). A framework for the study of persuasion. *Annual Review of Political Science*, 25(1):65–88.
- Esarey, J. and Menger, A. (2019). Practical and effective approaches to dealing with clustered data. *Political Science Research and Methods*, 7(3):541–559.
- Fraser, N. A. R. and Cheng, J. W. (2024). Do natives prefer white immigrants? evidence from Japan. *Ethnic and Racial Studies*, 47(5):977–1001.
- Gouda, M. and Song, J. (2024). The determinants of individual attitudes towards immigrants in south korea. *SSRN Electronic Journal*.
- Grigorieff, A., Roth, C., and Ubfal, D. (2020). Does information change attitudes toward immigrants? *Demography*, 57(3):1117–1143.
- Hainmueller, J. and Hopkins, D. J. (2014). Public attitudes toward immigration. *Annual Review of Political Science*, 17(1):225–249.
- Hainmueller, J. and Hopkins, D. J. (2015). The hidden american immigration consensus: A conjoint analysis of attitudes toward immigrants. *American Journal of Political Science*, 59(3):529–548.
- Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2014). Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments. *Political Analysis*, 22(1):1–30.
- Helbling, M., Maxwell, R., and Traunmüller, R. (2024). Numbers, selectivity, and rights: The conditional nature of immigration policy preferences. *Comparative Political Studies*, 57(2):254–286.
- Helbling, M. and Traunmüller, R. (2020). What is islamophobia? disentangling citizens’ feelings toward ethnicity, religion and religiosity using a survey experiment. *British Journal of Political Science*, 50(3):811–828.

- Hopkins, D. J., Sides, J., and Citrin, J. (2019). The muted consequences of correct information about immigration. *Journal of Politics*, 81(1):315–320.
- Hopkins, V., Lawlor, A., and Paquet, M. (2026). How do immigration policies affect voter support for low-skilled immigrants? evidence from a survey experiment. *International Migration Review*, 60(1):469–494.
- In, H. (2024). Ch’ulsanyul ollyödo ’irhal saram’ chunda... taehansangüi "imin munt’ök natch’wöya" [Even if the birthrate increases, the number of ’workers’ will still decrease... KCCI says "immigration barriers must be lowered"]. *Han’gugilbo*.
- Jin, S. (2025). Why is immigration important to you? a revisit to public issue salience and elite cues. *European Journal of Political Research*.
- Jung, D.-h. (2023). Economist challenges yoon’s immigration policies. *The Korea Times*.
- Kim, S. and Park, S. H. (2023). Living together with unease—koreans’ perception of and attitudes toward immigrants. In Kim, J., editor, *A Contemporary Portrait of Life in Korea: Researching Recent Social and Political Trends*, pages 25–50. Springer Nature, Singapore.
- Korea Herald (2025). Korea cuts e-9 visa quota to 80,000 for 2026.
- Korea Times (2025). South korea’s presidential race overlooks migrant issues. Accessed: 22 August 2025.
- Kustov, A. (2024). Beyond changing minds: Raising the issue importance of expanding legal immigration. *Perspectives on Politics*, pages 1–20.
- Kustov, A., Laaker, D., and Reller, C. (2021). The stability of immigration attitudes: Evidence and implications. *Journal of Politics*, 83(4):1478–1494.
- Leeper, T. J., Hobolt, S. B., and Tilley, J. (2020). Measuring subgroup preferences in conjoint experiments. *Political Analysis*, 28(2):207–221.

- Lenz, G. S. (2012). *Follow the Leader? How Voters Respond to Politicians' Policies and Performance*. University of Chicago Press, Chicago, IL.
- Letki, N., Walentek, D., Dinesen, P. T., and Liebe, U. (2024). Has the war in Ukraine changed Europeans' preferences on refugee policy? *Journal of European Public Policy*, 31(11):3764–3793.
- Levy, M. and Wright, M. (2020). *Immigration and the American Ethos*. Cambridge University Press, Cambridge.
- Liu-Farrer, G. (2020). *Immigrant Japan: Mobility and Belonging in an Ethno-nationalist Society*. Cornell University Press, Ithaca, NY.
- Maeda, K. and Strausz, M. (2025). Conservative politics and the dilemma of immigration in Japan. *Journal of East Asian Studies*, 25(3):352–367.
- Ministry of Justice (2024). Status of foreign residents in Korea.
- Nicholson, S. P. (2012). Polarizing cues. *American Journal of Political Science*, 56(1):52–66.
- Nyhan, B. and Reifler, J. (2010). When corrections fail: The persistence of political misperceptions. *Political Behavior*, 32(2):303–330.
- Park, B. (2024). 'top-tier' visa to be created to attract high-tech foreign workers. *Yonhap News Agency*.
- Seol, D.-H. and Skrentny, J. D. (2009). Ethnic return migration and hierarchical nationhood: Korean Chinese foreign workers in South Korea. *Ethnicities*, 9(2):147–174.
- Song, C. (2018). Joseonjok and goryeo saram ethnic return migrants in South Korea: Hierarchy among co-ethnics and ethnonational identity. In Tsuda, T. and Song, C., editors, *Diasporic Returns to the Ethnic Homeland*, pages 57–77. Palgrave Macmillan, Cham.
- Song, C. (2023). Joseonjok and goryeo saram ethnic return migrants in South Korea: Challenges of co-ethnic hierarchization and ethnonationalism. *Georgetown Journal of International Affairs*.

- Valentino, N. A., Soroka, S. N., Iyengar, S., Aalberg, T., Duch, R., Fraile, M., Hahn, K. S., Hansen, K. M., Harell, A., Helbling, M., Jackman, S. D., and Kobayashi, T. (2019). Economic and cultural drivers of immigrant support worldwide. *British Journal of Political Science*, 49(4):1201–1226.
- Wood, T. and Porter, E. (2019). The elusive backfire effect: Mass attitudes' steadfast factual adherence. *Political Behavior*, 41(1):135–163.
- Yu, S. (2023). Migrant racialization in South Korea: Class and nationality as the central narrative. *Ethnic and Racial Studies*, 46(10):2089–2110.

Supplementary Information for “Persuasion and Prejudice:
Are South Korean Attitudes Toward Immigration Open to
Change?”

Appendix A Additional Survey Information

The survey was fielded through Qualtrics’ online panel in January 2024, recording responses from 2,004 native-born South Koreans (1,999 after excluding respondents missing on raking dimensions). To ensure semi-representativeness, quotas were established in alignment with the most up-to-date demographic parameters from Statistics Korea (KOSIS, January 2024). The design included Qualtrics’ inbuilt quality controls (bot and duplicate detection), a Korean-citizen screen (Q6), and a manual attention check (Q7, “Please select strongly disagree”). Respondents who failed any of these checks exited the survey at the corresponding flow branch and were replaced through additional fielding. All replacement and quality-flag review occurred *before* the framing manipulation block in the Qualtrics flow. The citizen screen and the attention check sit on branches that terminate the response prior to the cue presentation, so no respondent who saw a framing cue was replaced post hoc. The 2,004 finished responses entering our analysis therefore all received a frame assignment and proceeded through every downstream block. To correct for over- and under-sampling, especially for education, we added post-stratification weights using the raking method (survey package in R) on age group, sex, education, and a five-category region (collapsing the seventeen administrative districts to capital, capital region, southeast, southwest, and other). Table A.1 reports unweighted sample counts and weighted proportions.

Variable	Count	Proportion	Weighted Prop.
Age			
18-29	357	17.9%	17.5%
30-39	348	17.4%	15.7%
40-49	397	19.9%	16.6%
50-59	529	26.5%	21.5%
60+	369	18.4%	28.6%
Sex			
Man	1,015	50.7%	50.0%
Woman	989	49.3%	50.0%
Education			
No University	388	19.4%	45.5%
University	1,616	80.6%	54.5%
Region (collapsed)			
Capital (Seoul)	873	43.7%	29.5%
Capital region (Gyeonggi, Incheon)	555	27.8%	20.9%
Southeast (Gyeongsang, Busan, Daegu, Ulsan)	264	13.2%	23.1%
Southwest (Jeolla, Gwangju)	103	5.2%	11.7%
Other (Chungcheong, Gangwon, Daejeon, Sejong, Jeju)	208	10.4%	14.8%

Table A.1: Summary counts, proportions, and weighted proportions for South Korea. Counts and un-weighted proportions are reported on the raw sample ($n = 2,004$ before exclusions for missing raking variables). Weighted proportions are computed on the analytic sample of $n = 1,999$. Small row-total deviations (age = 2,000, sex = 2,004) reflect item-level missingness on the demographic items and the raking-exclusion difference.

Sample derivation and participant flow

Frame assignment in the Qualtrics flow is implemented as an unconstrained uniform random integer drawn at survey entry (`RandomImmIntro = ${rand://int/1:3}`, FlowID FL_6), so eligibility, consent, and quality-assurance branches run on already-randomized respondents. Because the citizen screen, the attention check, and the bot and duplicate-detection branches all terminate the response *before* the framing manipulation block (FlowID FL_19), respondents removed at these checkpoints never reached a cue and were replaced through additional fielding rather than carried forward. The 2,004 finished re-

sponses in the export therefore correspond to respondents who passed all upstream quality controls and proceeded through every downstream block. The per-arm Ns at each stage of sample construction are listed below.

- Eligibility assessment covered 2,004 native-born Korean respondents who reached the framing block after passing Qualtrics screening and consent (upstream replacement of failed quality flags handled at fielding, not tracked at the respondent level in the export).
- Pre-randomization exclusions were 0. The random integer is assigned at survey entry. The upstream quality, citizen, and attention-check branches eliminate failing responses before the cue is presented, and those responses are replaced rather than reported.
- Random assignment covered 2,004 respondents total. Control = 666, Growth = 640, Birthrate (Fertility) = 698.
- The intended condition was received by 2,004 respondents, identical to the randomized count. The cue is a text screen presented immediately after assignment, so compliance loss is structurally zero in this online single-session design.
- Loss to follow-up was 0 in the single-session online survey with no inter-session continuation.
- Post-randomization exclusions were 5 total. Control = 2, Growth = 3, Birthrate = 0. All five drops are due to item-level missingness on one or more raking dimensions (four on age, one on region). None are due to the framing manipulation, the conjoint task, the attention check, or any post-treatment behavior.
- The framing experiment ITT analysis includes 1,999 respondents total. Control = 664, Growth = 637, Birthrate = 698. The same $n = 1,999$ enters the conjoint analysis, with profile-level rows clustered on respondent.

Baseline balance across framing arms

Table A.2 reports pre-treatment covariate distributions stratified by framing arm, together with the maximum pairwise standardized mean difference (SMD) across the three arm-pairs for each covariate. Contin-

uous covariates use a pooled-SD denominator. Binary indicators use $\sqrt{p_{\text{pool}}(1 - p_{\text{pool}})}$. Demographic balance is strong across age, sex, education, ideology, partisanship, and the pre-treatment sociotropic composite (all SMDs ≤ 0.09). The single covariate that exceeds the 0.10 balance threshold is the Southeast region (SMD = 0.16, with 11.0% Control, 16.5% Growth, 12.3% Birthrate), which falls within the variance expected from unconstrained random-integer assignment at $n \approx 660$ per arm and below the conventional 0.20 imbalance flag. The headline framing-experiment models in the main text condition on age group, sex, education, region, ideology, and partisanship, so the small Southeast imbalance is also adjusted away in estimation.

Covariate	Control	Growth	Birthrate	Max SMD
N (analyzed)	664	637	698	–
Age, mean (SD)	45.8 (14.5)	45.4 (14.6)	45.6 (14.6)	0.030
Female (%)	50.2%	48.8%	48.7%	0.029
University (%)	79.8%	80.2%	81.8%	0.050
<i>Region (5-category collapse)</i>				
Capital (Seoul)	46.4%	41.9%	42.3%	0.090
Capital region (Gyeonggi, Incheon)	28.2%	26.4%	28.5%	0.048
Southeast	11.0%	16.5%	12.3%	0.160
Southwest	5.3%	4.9%	5.3%	0.020
Other	9.2%	10.4%	11.6%	0.079
Ideology (1–10), mean (SD)	5.2 (1.8)	5.3 (1.9)	5.2 (1.8)	0.058
<i>Partisanship</i>				
Democratic Party (DP, %)	36.7%	35.0%	35.0%	0.037
People Power Party (PPP, %)	27.4%	29.4%	29.4%	0.043
Independent / Other (%)	35.8%	35.6%	35.7%	0.004
Sociotropic composite (1–7), mean (SD)	4.5 (1.1)	4.4 (1.1)	4.5 (1.1)	0.030

Table A.2: Pre-treatment covariate balance across framing arms. The SMD column reports the maximum pairwise standardized mean difference across the three arm-pairs. Continuous variables use a pooled-SD denominator; binary indicators use $\sqrt{p_{\text{pool}}(1 - p_{\text{pool}})}$. Values < 0.10 conventionally indicate balance; values ≥ 0.20 flag imbalance.

Appendix B Survey Questions

This section provides the text used for the background questions, the policy-preference dependent variable, the sociotropic battery, and the framing-cue messages.

Background Questions

- Sex at birth was recorded as Male / Female.
- Current residence was recorded as one of seventeen administrative regions.
- Age was recorded as a validated numeric input.
- Highest level of education was recorded as No formal / Elementary or below / Middle school / High school / Some college / University / Graduate / Other.
- Ideology was recorded on a 1-10 sliding scale (1 = very progressive, 10 = very conservative).
- Party preference was recorded for a hypothetical national election tomorrow as People Power Party / Minjoo Party / Green-Justice Party / Other Party / Don't know.

Sociotropic-Attitude Battery

The six-item battery used this prompt. "Please rate how much you agree with the following items (1 = strongly agree, 4 = neutral, 7 = strongly disagree)."

- The influx of foreign workers increases the competitiveness of the domestic agricultural industry.
- The influx of foreign workers financially secures pensions, stabilizes the population, and increases birth rates.
- The influx of foreign workers increases the competitiveness of the domestic financial sector.
- The influx of foreign workers is beneficial to Korea.
- The influx of foreign workers increases the competitiveness of domestic manufacturing.

- The influx of foreign workers is essential to domestic economic growth.

We reverse-code the items so that higher composite scores indicate more pro-immigration sentiment. Sociotropic optimists are coded as the top quartile of the composite (75th percentile cut, $n = 370$). The rest are skeptics ($n = 1,006$).

Framing Cue Messages

The following messages are verbatim from the instrument.

- *Control*: “Next, you will be asked your opinion about immigration. You will rate the immigrant profile as if you were an immigration officer.”
- *Economic Growth Cue*: “The government believes that the only way to maintain economic competitiveness and a high standard of living is to accept more immigrants, regardless of their background. Next, you’ll be asked for your opinion on immigration. You will rate immigrant profiles as if you were an immigration officer.”
- *Birthrate Cue*: “South Korea has one of the lowest fertility rates among advanced industrialized countries. The government believes that the only way to solve the economic and social problems caused by the low birth rate is to accept more immigrants, regardless of their background. Next, you will be asked for your opinion on immigration. You will rate immigrant profiles as if you were an immigration officer.”

The cue is reinforced after the fourth conjoint task with a shortened “Remember, the government believes ...” restatement.

Policy-Preference DV

The single-choice item asked, “In your view, should immigration levels be maintained at current levels, increased, or decreased?”

Conjoint Task Instructions

Verbatim English translation. “To pilot a new policy, the immigration office is reviewing applications from legal residents. You have been assigned to review applications under this pilot. You will see profiles of two immigrants who came to Korea. Of these two, please select the one you think is more suitable to be admitted.”

Appendix C Additional Figures and Tables

This section presents supplementary figures and tables underlying the headline polarization, co-ethnic preference, and attribute-hierarchy findings reported in the main text (Sections 4.1-4.3).

Figure C.1 reports subgroup results by sociotropic baseline orientation (optimists vs. skeptics). The growth cue produces a large consolidation effect in both groups. The fertility cue produces small shifts.

Figure C.2 presents marginal means for all conjoint attributes, including those not foregrounded in the main text (occupation, language, religion, marital status, age, years in Korea, employment history, Korea visit history, sex). The figure provides a complete picture of which attributes structure immigrant-admission preferences.

Tables C.1, C.2, and C.3 report regression-based estimates underlying the framing-experiment and origin-by-frame analyses in the main text.

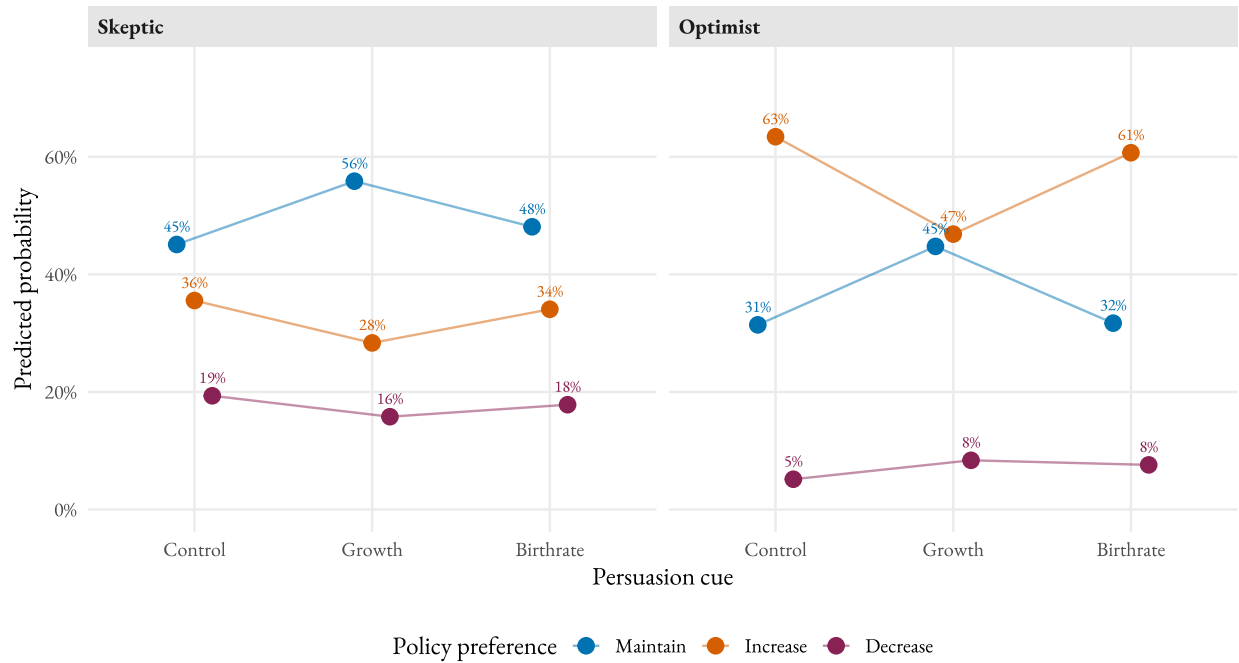


Figure C.1: Persuasion cues and immigration policy preferences, by sociotropic baseline orientation.

Note: Predicted probabilities from multinomial logistic regression with controls and survey weights. Optimists are the top quartile of the pre-treatment sociotropic composite (75th percentile cut). Skeptics are the rest. Within-arm n is optimists $\approx$ 110-130 and skeptics $\approx$ 230-260.

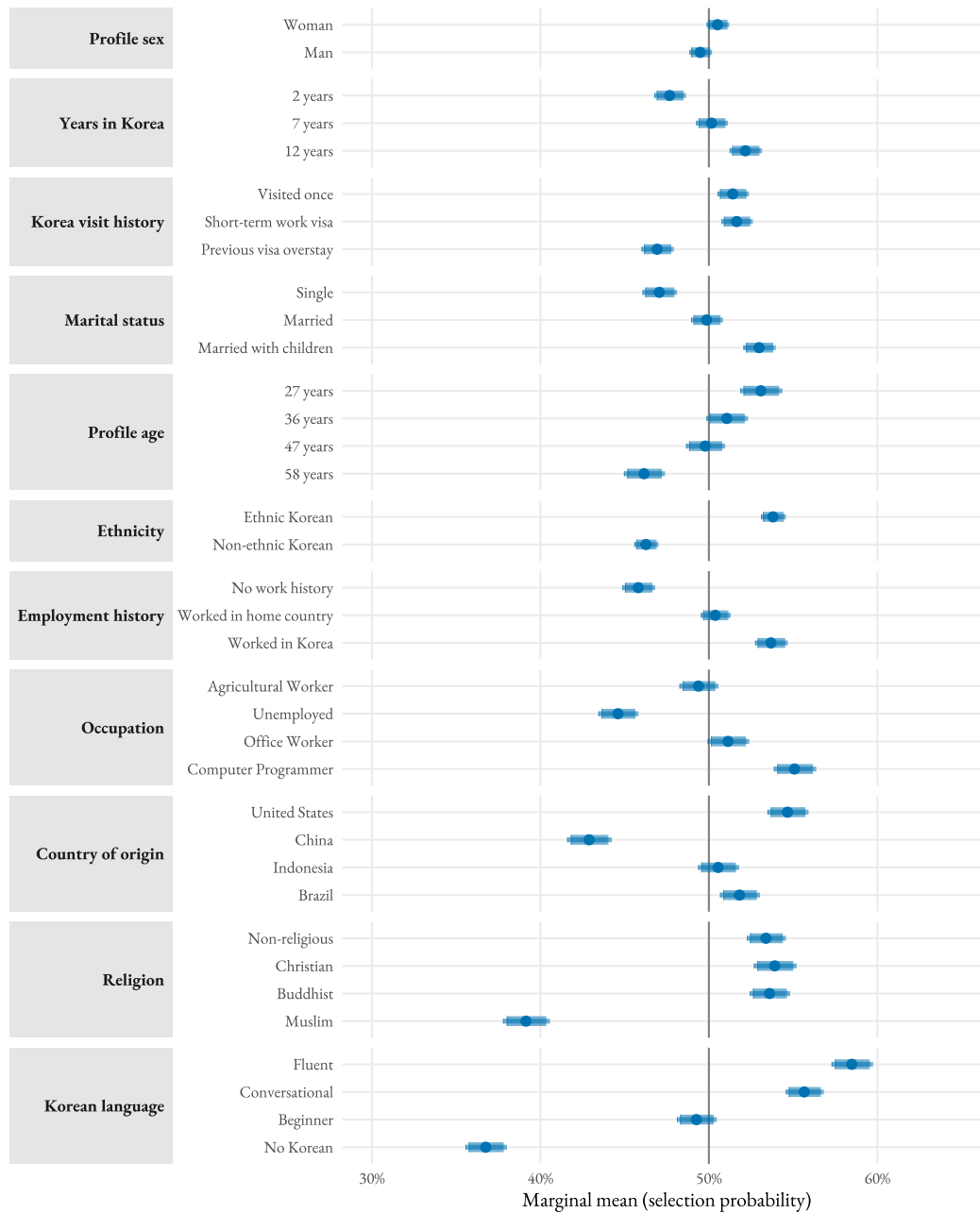


Figure C.2: Marginal means of all immigrant attribute levels, pooled across framing arms.

Note: 0.5 marks chance selection. Error bars show 90% (thick) and 95% (thin) confidence intervals with respondent-clustered standard errors and survey weights.

Outcome	Term	Estimate	SE	z	p
Decrease	frameBirthrate	0.0365	0.1553	0.235	0.8142
Decrease	frameGrowth	-0.1952	0.1590	-1.228	0.2196
Increase	frameBirthrate	-0.1216	0.1265	-0.961	0.3367
Increase	frameGrowth	-0.3131	0.1282	-2.442	0.0146

Table C.1: Multinomial logit treatment coefficients (relative to Maintain baseline). Controls: sex, age group, education, region, ideology, partisanship. Survey weights applied; cluster-robust SEs.

Variable	Estimate	SE	z-statistic	p-value
Support Increase				
Birthrate Cue	-0.0569	0.1484	-0.383	0.7015
Growth Cue	-0.1413	0.1533	-0.922	0.3566
Sociotropic Optimists	1.2567	0.1845	6.811	< 0.0001
Growth Cue $\times$ Optimists	-0.1769	0.2612	-0.677	0.4983
Birthrate Cue $\times$ Optimists	0.0972	0.2651	0.367	0.7139
Support Decrease				
Birthrate Cue	0.0615	0.1542	0.399	0.6900
Growth Cue	-0.1285	0.1610	-0.798	0.4248
Sociotropic Optimists	-1.4836	0.3140	-4.726	< 0.0001
Growth Cue $\times$ Optimists	-0.1637	0.4810	-0.340	0.7337
Birthrate Cue $\times$ Optimists	-0.8832	0.5782	-1.528	0.1265
Support Maintain				
Birthrate Cue	0.0017	0.1300	0.013	0.9896
Growth Cue	0.1936	0.1332	1.453	0.1463
Sociotropic Optimists	-0.5354	0.1804	-2.967	0.0030
Growth Cue $\times$ Optimists	0.2004	0.2519	0.796	0.4262
Birthrate Cue $\times$ Optimists	0.0760	0.2581	0.294	0.7686

Table C.2: Multinomial logit treatment effects by sociotropic orientation

Note: Reference categories are Control treatment and Sociotropic Skeptics. Survey weights applied. Cluster-robust SEs.

Treatment Contrast	Origin Contrast	Estimate	SE	t-statistic	p-value
Growth – Control					
Growth – Control	China – USA	-0.026	0.025	-1.04	0.298
Growth – Control	Indonesia – USA	0.030	0.023	1.30	0.193
Growth – Control	Brazil – USA	0.006	0.022	0.27	0.787
Birthrate – Control					
Birthrate – Control	China – USA	-0.005	0.024	-0.21	0.834
Birthrate – Control	Indonesia – USA	0.028	0.023	1.22	0.224
Birthrate – Control	Brazil – USA	0.033	0.022	1.50	0.134

Table C.3: Treatment $\times$ Origin AMCE differences (Cue – Control), pooled.

Note: AMCE differences for each origin relative to USA, computed as (Cue - Control). Weighted OLS with respondent-clustered SEs.

Appendix D Power, Minimum Detectable Effect, and Equivalence Tests

The achieved sample is $n = 1,999$ native-born South Korean respondents recruited through Qualtrics in January 2024, with random assignment to one of three framing arms (control, growth, fertility) producing arm sizes of approximately 660 to 700 respondents. The conjoint module presented ten paired-profile tasks per respondent, yielding approximately 39,980 profile-level observations clustered within 1,999 respondents (about 13,300 per arm). We treat the respondent as the unit of clustering and compute robust standard errors accordingly (Esarey and Menger, 2019).

To benchmark the precision of our estimates, we compute minimum detectable effects (MDEs) on the percentage-point scale for two-sided tests at $\alpha = 0.05$ and 80% power, using the closed-form expressions in Schuessler and Freitag (2020). The within-respondent intraclass correlation in the conjoint outcome (forced choice) is approximately zero by design (each respondent has exactly five chosen and five unchosen profiles), so the design effect is approximately 1. Table D.1 reports MDEs for the principal estimand classes.

Estimand class	Profiles per cell	MDE (pp)
Pooled main-effect AMCE (e.g., Ethnicity)	19,990	1.40
Within-arm pooled AMCE	3,332	3.43
Treatment x origin AMCE difference	3,332	4.85
Treatment x origin x ethnicity (3-way) AMCE difference	1,666	6.86
Treatment x origin x skill (3-way)	1,666	6.86

Table D.1: Minimum detectable effect (MDE) by estimand class.

Note: Two-sided $\alpha = 0.05$, power = 0.8. MDEs use the achieved sample, the within-respondent intraclass correlation estimated from the data ($\rho \approx 0$), and the implied design effect ($DEFF \approx 1$). The within-arm and three-way MDEs are the pooled MDE inflated by $\sqrt{2}$ for the relevant contrast.

Several of the directional effects highlighted in the manuscript fall below these MDEs. The fertility-cue modifications of the low-skill origin penalty (Indonesia +2.3, Brazil +5.3, China −5.3 percentage points) are substantively interesting but lie around or beneath the 4.85-point MDE for treatment-by-

origin interactions. Reporting them as null effects under conventional null-hypothesis significance testing (NHST) is therefore misleading. The design lacks the precision to distinguish these from a true effect in the 1.5- to 5-point range. Equivalence testing with a justified smallest effect size of interest (SESOI) clarifies that the evidence supports the absence of *large* effects rather than the absence of effects (Lakens, 2017; Rainey, 2014).

We adopt a SESOI of five percentage points for treatment-by-origin interactions and policy-preference shifts. Five points represents one-third of the largest origin penalty observed in the experiment (the China-versus-U.S. AMCE of approximately 10 points) and approximates the smallest shift that would alter South Korea’s stratified visa hierarchy in observable terms. An EPS quota adjustment of comparable magnitude affects roughly twenty thousand workers per year. Following Lakens et al. (2018), we use the two one-sided tests procedure. Each effect estimate is tested against -0.05 and $+0.05$, and equivalence to zero (within the SESOI band) is concluded only when both one-sided tests reject. We report the upper bound of the 90% confidence interval for each effect as the effect-size ceiling, following Rainey (2014).

Table D.2 reports the equivalence-test conclusions for the framing-experiment policy-preference effects (full numerical output is generated by the analysis pipeline).

Contrast	Estimate (90% CI)	Equivalence to zero (SESOI ± 5 pp)
Growth – Control on Maintain	+6.5 [0.5, 11.9]	Not equivalent. Substantive consolidation.
Growth – Control on Increase	−5.8 [−11.2, 0.0]	Not equivalent. Consolidation pulls from Increase.
Growth – Control on Decrease	−0.8 [−5.2, 3.3]	<i>Equivalent</i> within the SESOI band.
Birthrate – Control on Maintain	+1.9 [−3.6, 7.9]	Not equivalent. Underpowered null.
Birthrate – Control on Increase	−3.0 [−8.8, 2.2]	Not equivalent. Underpowered null.
Birthrate – Control on Decrease	+1.1 [−3.2, 5.7]	Not equivalent. Underpowered null.

Table D.2: Equivalence-test conclusions for framing-experiment policy-preference effects.

Note: Estimates are percentage-point shifts from the control arm. “Equivalent” means both one-sided tests against ± 5 pp reject. “Not equivalent” means at least one one-sided test does not reject and the data cannot rule out a SESOI-sized effect.

The conclusion is that we have substantive evidence of consolidation under the growth cue, while the fertility-cue effects on direction are too imprecise to characterize as substantively zero. The data are consistent with anything from a small liberalizing fertility-cue effect to a small consolidation effect, but rule out a large effect in either direction. For the conjoint, treatment-by-origin AMCE differences for the China contrast are similarly underpowered. The upper bound of the 90% CI on the Growth-Control China-USA contrast is +0.015 (1.5 percentage points), well within the SESOI. The upper bound for the Birthrate-Control China-USA contrast is +0.029 (2.9 percentage points). Larger origin penalties (the 10-point China main effect, the 7-point ethnicity effect, the 14-point Muslim penalty) are all detected with high power, well above the MDEs in Table D.1.

Appendix E Sociotropic Orientation and the Attribute Hierarchy

The main text reports that the headline attribute hierarchies (origin, religion, language, ethnicity, skill) are stable across framing arms. A natural follow-up is whether they are also stable across respondents' baseline sociotropic orientation. Figure E.1 reports marginal means for origin, ethnicity, religion, and occupation, split by the top-quartile sociotropic optimists ($n = 370$) and the rest ($n = 1,006$ skeptics). Optimists are, by construction, the most pro-immigration respondents on the abstract battery. Their attribute hierarchies are similar to the skeptics'. The China-versus-USA origin penalty, the non-Korean ethnicity penalty, the anti-Muslim premium, and the skill premium all reappear at the same rank order in both groups, with magnitudes overlapping within the 95% bands. Pro-immigration sociotropic reasoning shifts levels. It leaves the structure intact because respondents whose general orientation is open continue to evaluate specific candidates through the same prior commitments.

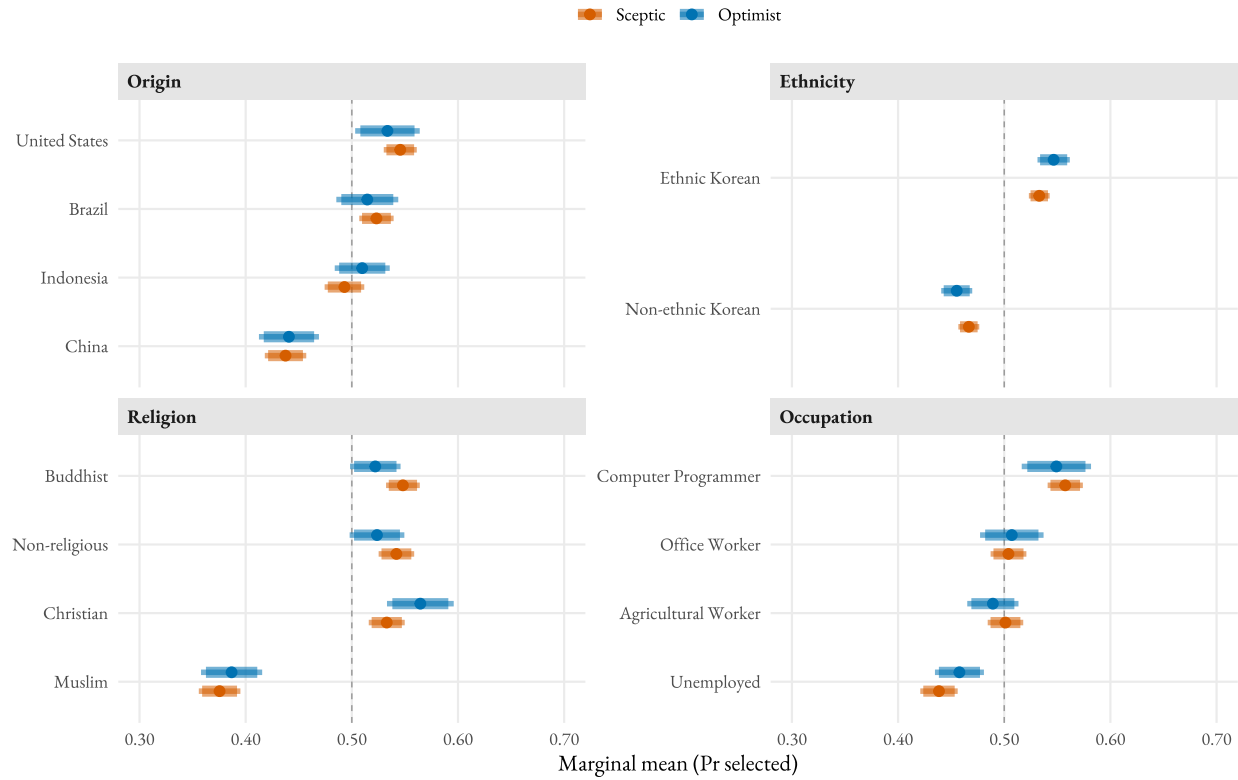


Figure E.1: Conjoint marginal means by sociotropic group. Prior commitments persist among the most pro-immigration respondents.

Note: Marginal means with respondent-clustered standard errors and survey weights. Sociotropic optimists are the top quartile of the pre-treatment sociotropic composite ($n = 370$). Sceptics are the rest ($n = 1,006$). Thick lines are 90% confidence intervals. Thin lines are 95%. The dashed vertical line at 0.5 marks chance selection.

Appendix F Statistical Significance Summary

Table F.1 consolidates every headline statistical test reported in the main text. The first four blocks are NHST tests where small p -values indicate a detectable effect. The equivalence-test block (against the $\pm 5\text{pp}$ SESOI of Appendix D) is the inverse. Small p -values mean the effect is bounded inside the SESOI band.

Test	Estimate	p	Sig.
<i>Pooled framing effects on policy preference (multinomial logit, vs. Maintain)</i>			
Growth — Increase	-0.313	0.015	*
Birthrate — Increase	-0.122	0.337	
Growth — Decrease	-0.195	0.220	
Birthrate — Decrease	+0.037	0.814	
<i>Polarization (LPM): Cue × Ideology interactions</i>			
Growth × Centrist on Maintain	+20.6	0.002	**
Birthrate × Centrist on Maintain	+4.3	0.507	
Growth × Conservative on Maintain	+26.2	< .001	***
Birthrate × Conservative on Maintain	+7.4	0.356	
Growth × Centrist on Increase	-7.6	0.225	
Birthrate × Centrist on Increase	+2.2	0.720	
Growth × Conservative on Increase	-11.0	0.133	
Birthrate × Conservative on Increase	+1.4	0.852	
Growth × Centrist on Decrease	-13.0	0.010	*
Birthrate × Centrist on Decrease	-6.5	0.192	
Growth × Conservative on Decrease	-15.2	0.011	*
Birthrate × Conservative on Decrease	-8.8	0.152	
<i>Polarization (LPM): Cue × Partisanship interactions</i>			
Growth × Independent on Maintain	-5.2	0.422	
Birthrate × Independent on Maintain	-4.8	0.449	
Growth × PPP on Maintain	+14.0	0.041	*
Birthrate × PPP on Maintain	+1.1	0.872	
Growth × Independent on Increase	+9.9	0.103	
Birthrate × Independent on Increase	+5.5	0.357	
Growth × PPP on Increase	-3.0	0.639	
Birthrate × PPP on Increase	-0.3	0.963	
Growth × Independent on Decrease	-4.7	0.341	
Birthrate × Independent on Decrease	-0.7	0.890	
Growth × PPP on Decrease	-11.0	0.036	*
Birthrate × PPP on Decrease	-0.8	0.878	
<i>Co-ethnic preference (LPM on chosen)</i>			
Birthrate vs. Control, Ethnic-Korean Unemployed profiles	+5.6	< .001	***
Birthrate vs. Control, low-skill Brazil profiles	+5.3	0.002	**
Birthrate vs. Control, low-skill China profiles	-5.3	0.001	**
<i>Equivalence tests (TOST, ±5pp SESOI)</i>			
Growth vs Control, Maintain	+6.5	0.671	
Birthrate vs Control, Maintain	+1.9	0.190	
Growth vs Control, Increase	-5.8	0.593	
Birthrate vs Control, Increase	-3.0	0.280	
Growth vs Control, Decrease	-0.8	0.050	†
Birthrate vs Control, Decrease	+1.1	0.080	

Table F.1: Summary of significance tests for all headline estimates reported in the main text.

Note: Pooled framing-effect estimates are multinomial-logit coefficients (relative to Maintain) and are expressed in log-odds; all other estimates are linear-probability-model coefficients in percentage points. Polarization interaction estimates are the Growth-cue (or Birthrate-cue) interaction with the listed subgroup category relative to the omitted Progressive (ideology) or Democratic-Party (partisanship) reference, on the listed outcome. Co-ethnic-preference LPMs control for all other conjoint attributes; survey weights applied; respondent-clustered SEs. The equivalence tests block reports the two-one-sided-test (TOST) p -value against a ± 5 pp smallest effect size of interest. Significance codes: † $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$. For the equivalence-test rows, the † symbol marks rows where both one-sided tests reject (i.e., the effect is statistically equivalent to zero within the SESOI band).

Appendix G Conjoint Diagnostics and Robustness Checks

This appendix reports four standard conjoint diagnostics as robustness checks on possible design artifacts in the headline results: profile-position effects, task-fatigue, attention-pass sensitivity, and cue-reinforcement.

G.1 Profile-position effect

A position effect would arise if respondents preferred the left- or right-hand profile irrespective of attribute content. Table G.1 reports the mean selection rate by profile position and the cluster-robust contrast.

Table G.1: Profile-position effect (left vs right).

Position	Selection rate	N profiles
Left (Profile A)	50.3%	19990
Right (Profile B)	49.7%	19990
Right – Left (pp)	-0.52	
90% CI	[-1.50, +0.46]	
<i>p</i> -value	0.302	

The selection rate is 50.3% for Profile A (left) and 49.7% for Profile B (right). The contrast is -0.52 percentage points (90% CI $[-1.50, +0.46]$, $p = .30$) and is indistinguishable from zero. Respondents allocate attention symmetrically across the two positions.

G.2 Task-fatigue

Respondents complete ten paired tasks. If late-task responses are less attentive or anchored, attribute coefficients should drift with task number. Table G.2 reports the origin AMCE (against the United States baseline) computed separately on three task blocks: tasks 1–3, tasks 4–6, and tasks 7–10.

Table G.2: Origin AMCE (vs. United States) by task block. Estimates on the probability scale with respondent-clustered standard errors in parentheses.

Origin	Tasks 1-3	Tasks 4-6	Tasks 7-10
China	-0.097 (0.018)	-0.109 (0.018)	-0.140 (0.016)
Indonesia	-0.018 (0.017)	-0.068 (0.017)	-0.039 (0.016)
Brazil	-0.004 (0.018)	-0.047 (0.017)	-0.033 (0.013)

The China penalty is larger in each successive block (-9.7pp in tasks 1–3 and -14.0pp in tasks 7–10), and several origin-by-task-block interactions reach conventional significance levels (e.g., Indonesia $\times$ Tasks 4–6, $p = .006$). The pattern is consistent with respondents *intensifying* rather than disengaging from the discrimination structure as the task block progresses. We do not read this as fatigue in the disengagement sense, which would produce noisier, smaller estimates. It does indicate that any test using only early-task responses would understate origin penalties.

G.3 Attention-pass sensitivity

The instrument incorporated a manual attention check (“Please select strongly disagree”, Qualtrics column Q7 . . . 28). The check was a hard gate at the Qualtrics layer, and 100% of finished respondents in the export passed. The analytical sample of $n = 1,999$ further drops five respondents missing on a raking dimension. To probe sensitivity, Table G.3 compares headline polarization estimates on the analytical sample to the broader attention-pass sample of $n = 2,001$ (the five non-raking respondents plus three with missing ideology data drop here too).

Table G.3: Headline polarization estimates on the main analytical sample vs. a broader attention-pass sample (LPM, unweighted).

Sample	Estimand	Estimate (pp)	SE (pp)	p	n
Main ($n = 1,999$)	Growth $\times$ Centrist on Maintain	+26.71	6.64	< 0.001	1,999
Main ($n = 1,999$)	Growth $\times$ Conservative on Maintain	+27.63	8.17	< 0.001	1,999
Broad ($n = 2,001$)	Growth $\times$ Centrist on Maintain	+26.51	6.63	< 0.001	2,001
Broad ($n = 2,001$)	Growth $\times$ Conservative on Maintain	+27.78	8.16	< 0.001	2,001

The interaction estimates differ by 0.2 percentage points or less across the two samples. The main analyses are insensitive to whether the five raking-incomplete respondents are retained. Because all retained respondents passed the manual attention check at fielding, the analyses do not require a further attention-screen.

G.4 Cue-Reinforcement Before and After Reinforcement

By design the framing cue is reinforced after the third conjoint task to mitigate fade-out. Table G.4 reports the origin AMCE separately for the pre-reinforcement block (tasks 1–3) and the post-reinforcement block (tasks 4–10).

Table G.4: Origin AMCE (vs. United States) by reinforcement stage and framing arm. Percentage points (clustered SE in parentheses).

Origin	Post (tasks 4–10) — Control	Pre (tasks 1–3) — Control	Post (tasks 4–10) — Growth	Pre (tasks 1–3) — Growth	Post (tasks 4–10) — Birthrate	Pre (tasks 1–3) — Birthrate
China	-12.34 (2.04)	-6.76 (3.08)	-13.38 (2.42)	-12.01 (3.21)	-12.34 (2.07)	-10.40 (3.07)
Indonesia	-9.51 (1.94)	+1.51 (2.82)	-2.93 (2.04)	-1.89 (2.99)	-3.21 (1.98)	-4.78 (2.87)
Brazil	-5.43 (1.94)	-2.29 (3.23)	-3.29 (1.79)	-2.04 (3.28)	-3.01 (1.81)	+3.07 (2.94)

The post-reinforcement origin penalties match or exceed the pre-reinforcement estimates in every cell. Under the Control arm the China penalty roughly doubles, from -6.8pp in tasks 1–3 to -12.3pp in tasks 4–10 (delta -5.6pp , $p = .13$), and the Indonesia penalty changes from $+1.5\text{pp}$ to -9.5pp (delta -11.0pp , $p = .003$). Under the Growth and Birthrate cues the post-pre deltas are smaller and the

post-cue estimates closely track the corresponding cued estimates from the pre-block. Read together, the diagnostic suggests that the cue-and-reinforcement design stabilizes the cued discrimination structure across the conjoint sequence, while the Control-arm trajectory reflects ordinary within-task learning by respondents about how to discriminate among admission profiles.

References

- Esarey, J. and Menger, A. (2019). Practical and effective approaches to dealing with clustered data. *Political Science Research and Methods*, 7(3):541–559.
- Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362.
- Lakens, D., Scheel, A. M., and Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2):259–269.
- Rainey, C. (2014). Arguing for a negligible effect. *American Journal of Political Science*, 58(4):1083–1091.
- Schuessler, J. and Freitag, M. (2020). Power analysis for conjoint experiments. *SocArXiv*.